BEYOND
EVALUATION
USE

An Empirical Study of
Involvement and Use/Influence in
Multi-Site Evaluations

Report 4:
A Case Study of the Impact of the Local Systemic Change through Teacher Enhancement (LSC) Core Evaluation

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Last Updated: June 23, 2009

Note: This material is based upon work supported by the National Science Foundation under Grant No. REC 0438545. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

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Executive Summary

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the STEM education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of science, technology, engineering, and mathematics (STEM) education and evaluation were secondary audiences, not directly the intended users.

Evaluation projects related to the following four NSF programs were included in the study: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), Local Systemic Change through Teacher Enhancement (LSC), and Mathematics and Science Partnerships – Research, Evaluation and Technical Assistance (MSP-RETA).

Data were collected through an online survey distributed to evaluation project leaders[1] and evaluators, interviews with project leaders and evaluators, an interview with one of the NSF program officers responsible for the program at the time, an on line survey of NSF education project PIs and questions of journal editors. Additional data were collected through review of archival documents, publications, and reports produced by the evaluation projects and through reflective descriptions provided by the evaluation projects principal investigators.

This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the LSC program evaluation project. The case study showed a modest relationship between project involvement in the LSC core evaluation and the influence of the evaluation's processes and products on the projects. It appeared from the surveys and interviews with project personnel that perceptions of involvement were positively linked to perceptions of use. Other variables affecting use included where the project was in its life cycle, the perceived relevance of the core evaluation to the local project, and interactions with NSF.

The LSC core evaluation had only a small discernable effect on the fields of STEM education and evaluation. This effect was probably diffused and not directly attributable to the LSC evaluation. The analysis of the instances where LSC core evaluation products were cited in scholarly research showed that there was some use, particularly of the instruments, by people in LSC projects and elsewhere. In addition, the survey of NSF PIs revealed that the respondents had little knowledge of evaluations in general and practically no knowledge of the LSC evaluation.

In summary, the LSC core evaluation had a moderate level of influence on the LSC projects, which was enhanced by the involvement of the projects in the evaluation. The effect on the fields of STEM education and evaluation was much less apparent in the data collection conducted for this case study.

Introduction

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the STEM education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of STEM education and evaluation were secondary audiences, not directly the intended users.

Evaluation projects related to the following four NSF programs were included in the study: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), Local Systemic Change through Teacher Enhancement (LSC), and Mathematics and Science Partnerships – Research, Evaluation and Technical Assistance (MSP-RETA).

This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the LSC program evaluation project. The case study has three sections. First, it describes the history of the LSC core evaluation and the context within which the evaluation was conducted. Second, it examines the relationship between project-level involvement in the evaluation and project use of the evaluation processes and outcomes. And finally, it describes the extent to which the LSC core evaluation influenced the fields of STEM education and evaluation.

Methods

Data were collected through an online survey distributed to LSC project leaders[2] and evaluators, interviews with LSC project leaders and evaluators, an interview with one of the NSF program officers responsible for the LSC program at the time, an online survey of NSF education project Principal Investigators (PIs), and a short survey of journal editors. Additional data were collected through review of archival documents, publications, and reports produced by the LSC program and core evaluation and through reflective descriptions provided by the LSC core evaluation PI.

LSC Project Leader and Evaluator Survey

The research team acquired names and email addresses for the PIs, co-PIs, and evaluators of LSC projects. The team compiled lists from project and meeting rosters provided by the program evaluation personnel. When email addresses were not provided for individuals on the

list, a member of the team searched the Internet for an email address. In all, the research team acquired names and email addresses for a total of 139 LSC representatives.

Each individual received an initial invitation email with a description of the survey, an explanation of its use, the Human Subjects approval number, and telephone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and four weeks following the first email, the team sent email reminders to nonrespondents. The data were collected from August 10, 2006 through January 17, 2007.

Complete details of the survey's development, administration, and validation are presented in "Beyond Evaluation Use Report 1: NSF Follow-up Survey Detailed Data Report." Survey findings presented in this report are taken from 57 closed-ended items and four open-ended questions related to the involvement in and use of the LSC program evaluation.

A total of 74 surveys were completed by LSC project leaders and evaluators for an overall response rate of 53%. However, one of the respondents provided data deemed unusable, so that survey was removed from the analysis file, reducing the total number of LSC respondents to 73. Of the 73 respondents, 43 were LSC project leaders and 30 were LSC evaluators.

Table 1 shows the four open-ended items and the number and percentage of respondents who provided comments for each one.

Table 1. Responses to open-ended items

Type of Comment	Comment[3]	No Comment	Percent Respondents with Comments
Example of factors that enabled use	52	22	70.3%
Example of factors that limited use	46	28	62.2%
General comments about the program evaluation	38	36	51.4%
Example of a negative influence of the program evaluation	11	63	14.9%
Example of inappropriate use	3	71	4.1%

Nonrespondent Study. To check for nonresponse bias, the team conducted a nonrespondent follow-up survey after the online survey of the PIs and evaluators. We developed a three-question nonrespondent survey to address the topics of involvement and use, as well as to gather some information about what kept individuals from responding to the initial survey and reminder emails. The nonrespondent survey items were:

1. How involved were you in the ATE Program Evaluation?
2. How much impact did the ATE Program Evaluation have on you?
3. What was the main thing that kept you from responding to our initial survey request?

A random sample of nonrespondents to the survey was drawn using SPSS. The team sent each person an individually addressed email with a request for input on the three items. Two reminder emails were sent to each of the respondents who did not respond to the initial request. Then, if email contact did not solicit a response, a team member contacted the individual by telephone. Forty-nine percent of the nonrespondents (36/74) completed the nonrespondent survey.

The results of the analysis comparing the LSC respondents and nonrespondents showed no significant difference in the overall level of involvement reported (p=.147) or in the overall level of impact (p=.703). Means of the survey's involvement and use items were used to measure level of involvement and impact for respondents for comparison to the nonrespondents (see Table 2). The process of distilling the larger survey's multiple items related to use into a single question resulted in the choice of the word "impact" as a global concept to encompass questions about use of the process, instruments, and results of the evaluation. Therefore, the difference between respondents and nonrespondents may indicate the possibility of upward bias in the estimates of overall use for ATE survey respondents or it may also be an artifact of the difference between use, influence and overall impact.

Table 2. Results of nonrespondent study

	Involvement Mean			**Use/Impact Mean**		
	Respondents	Non-Respondents	p-value	Respondents	Non-Respondents	p-value
LSC	2.71	3.14	0.147	2.85	2.69	.703

Interviews

Research team members conducted in-depth interviews with a purposively selected sample of four project leaders and evaluators who completed the online survey (the Project Leader and Evaluator survey) and said they were willing to be interviewed. Survey respondents' responses to the involvement and use questions were used to identify four key types of projects: 1) high involvement, high use; 2) high involvement, low use; 3) low involvement, high use; and 4) low involvement, low use.

These categories were determined based on relative reported levels of involvement and use, with a mean response of 2.5 on the use and involvement questions as the approximate cut-off point between high and low levels. Interview participants were selected based not only on their responses to the survey but also to balance project leaders and evaluators and ensure diversity in the number of projects sampled.

In the second round, individuals who reported high use of the evaluation were invited to be interviewed. In the end, three interviews with "high users" were conducted. Again the

interviews were tape recorded and transcribed. Table 3 provides a description of the eight individuals interviewed in this study in terms of their levels of use and involvement and their status as evaluators, PIs, or other staff members.

Table 3. Description of interviewees

	Low Use	**High Use**
Low Involvement	Interviewee 4 (Evaluator)	Interviewee 1 (PI) Interviewee 6 (Evaluator)
High Involvement	Interviewee 3 (Evaluator)	Interviewee 2 (PI) Interviewee 5 (Evaluator) Interviewee 7 (Evaluator)

Frances Lawrenz, PI for the University of Minnesota project, conducted a telephone interview with one of the NSF program officers responsible for the LSC program. This interview was approximately 25 minutes long and was taped and transcribed.

Citation Analysis

The citation analysis used bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work. Web of Science, Google Scholar, and Google were used to collect citation information from lists of evaluation products produced by the LSC program evaluation (see Appendices A and B for lists of the evaluation products and citing works).

The list of the LSC evaluation products was initially compiled by reviewing the Horizon Research Inc. (HRI) LSC core evaluation website http://www.horizon-research.com/LSC/news. The list was then forwarded to HRI staff members who reviewed the list and made changes. The LSC core evaluation principal investigator was then provided with the opportunity to review the final list. Products were defined as any publications, presentations, instruments, or other related materials that were produced as part of the NSF-funded evaluation project.

The Web of Science, Google Scholar, and Google search engines were used to collect citation information from the evaluation product list. Searching was completed in late 2007 during the following periods: Web of Science October 23-25, Google Scholar October 25-November 13, and Google November 13 – December 2.

Data were collected manually by entering full reference information about the citations into an Excel spreadsheet which was later checked for accuracy and compliance with APA style and converted into an SPSS datafile. Complete details of the rationale, development, administration, and validation of the citation analysis are presented in “Beyond Evaluation Use:

A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields, Report 6."

NSF PI Survey

As a measure of influence on the field, the research team administered an online survey to a sample of 350 individuals randomly selected from the database of over 2,000 currently-funded NSF Education and Human Development Directorate Principal Investigators (PIs). The survey elicited responses about PIs' perceptions of evaluations, people, and publications that might have been influential on the fields of STEM evaluation and education. The overall response rate was 54.7%. Complete details of the survey's development, administration, and analysis are presented in "Beyond Evaluation Use Report 7: Findings from the Survey of Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field."

Journal Editor Inquiry

A very brief email/telephone survey of editors and coeditors of six journals in the evaluation and STEM education fields was also conducted to determine if the editors were aware of any articles that had been published about influential STEM evaluations. Responses were obtained from three of six editors/coeditors. Details about this inquiry are also included Report 7.

Archival Review

The collection of archival documents for LSC was primarily gathered using electronic databases and search engines. The NSF website provided information on the LSC program overall, as well as on individual LSC projects. The archival documents included reports, grant products, survey instruments, journal articles, NSF grant solicitations, project summaries, and newsletters.

Another major source of electronic information was the LSC core evaluation homepage at Horizon Research Inc. (http://www.horizon-research.com/LSC), as well as the Local Systemic Change Network (http://lsc-net.terc.edu/go.cfm/home) which is an electronic community of LSC PIs, project directors, evaluators, and other senior staff of LSC projects nationwide. This site is also known as LSC-Net and was designed to facilitate sharing of best practices, issues, and resources among LSC projects. In addition to links to a number of individual project websites, these two sites provided access to data collection protocols, reports, PowerPoint presentations, articles, and other products.

We also searched the Internet using Google, Google Scholar, and several electronic indexes including Academic Search, WilsonWeb (the H.W. Wilson Information Retrieval System for the World Wide Web), ERIC (the Education Resources Information Center), and the NSF-sponsored OERL (the Online Evaluation Resource Library). Complete details of the document collection and analysis are presented in "Beyond Evaluation Use Report 6: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields.

LSC Program and Core Evaluation Background and Context

This section of the case study presents details of both the LSC Program and the core evaluation to provide context for the discussion of evaluation involvement and use.

LSC Program

The National Science Foundation (NSF) launched the Local Systemic Change through Teacher Enhancement initiative (LSC) in 1995 to improve K-12 science, mathematics, and technology instruction through teacher professional development. LSC intended to shift the professional development focus from individual teachers to district- or system-wide populations of teachers, based on the premise that system-wide initiatives yielded better results.[4]

LSC Program Facts

Period: 1995 – 2005 (Final year of new projects were funded was 2002.)

Purpose: Improve instruction in science, mathematics and technology through teacher professional development within entire school districts.

Objectives: Target all teachers in a jurisdiction for professional development, requiring a minimum of 130 hours of professional development; and prepare teachers to implement district-designated instructional materials.

Primary Target: K-12 teachers of science and mathematics; focus on entire school systems or districts, not on individual teachers.

Sites Funded: 88 projects current and completed projects across 31 states, and involving 70,000 teachers, 4,000 schools, 467 school districts.

The LSC was a $250 million program that funded 88 projects, reaching 4,000 schools across 31 states and 476 districts in the period 1995-2005. Some LSC projects took place in single school district and others covered multiple districts (up to 20 in some cases). Half of the schools were urban, a quarter suburban, and the remaining quarter split between small cities and rural communities. LSC-treated teachers taught approximately 70,000 students, approximately half of whom were from minority groups. The professional development sessions took place both in the summer as well as throughout the academic year, providing teachers with opportunities to develop and apply new knowledge and skills within a supportive school/district culture.[5]

LSC Core Evaluation Background

In 1995, NSF contracted with Horizon Research, Inc. (HRI) and PI Iris Weiss, Ph.D., to conduct the nationwide LSC core evaluation.[6] NSF required LSC projects both to conduct individual project-level evaluations and to participate in the national cross-project core evaluation.

The core evaluation was intended to assess the status of individual LSC projects, as well as to collect information from as many diverse projects that could be aggregated to inform broader messages to NSF, Congress and the public. NSF's theory of action was that high quality professional development focused on standards-based instructional materials would improve teachers' knowledge and skills and would increase their disposition to use those materials in their classrooms.

Although somewhat modified over time as NSF's information needs changed and the need for longitudinal data became apparent, the core evaluation required each project to:

- Observe 5-8 professional development sessions per year and record observations on standardized protocols;
- Administer 300 teacher questionnaires to a random sample of teachers in the participating school districts;
- Administer questionnaires to principals in all participating schools;
- Conduct classroom observations of a minimum of 16 randomly selected teachers;
- Conduct interviews with 10 randomly selected teachers;
- Interview the project administrative team; and
- Collect information on district policies and practices.

LCS Core Evaluation Facts

NSF Contract Amount
- $6.25 million

NSF Goals for LSC Core Evaluation
- Provide aggregated information across diverse projects to inform broader conclusions about design, quality, and impact of LSC projects
- Assess individual projects to provide for mid-course adjustments

NSF Contracted with HRI to:
- Develop a data collection framework
- Provide technical assistance in implementing evaluation activities
- Prepare cross-site analyses of evaluation results

Local Project Evaluator Activities:
- Professional Development Session Observations (2,40 completed)
- Classroom Observations of mathematics and science lessons (1,620)
- Teacher Questionnaires (75,000)
- Principal Questionnaires (17,380)
- Teacher Interviews (1,782).

Initially, NSF did not require the projects to track student outcome data, but some chose to do it on their own. The collection of student achievement data was revisited later in the program and the decision was made to require newly-funded projects to assess impact on students starting.

LSC Core Evaluation Questions

The core evaluation was designed to address the following questions:

1. What is the overall quality of the LSC professional development activities?
2. What is the extent of school and teacher involvement in LSC activities?
3. What is the impact of the LSC professional development on teacher preparedness, attitudes, and beliefs about science and mathematics teaching and learning?

4. What is the impact of the LSC professional development on classroom practices in science and mathematics?
5. To what extent are the school and district contexts becoming more supportive of the LSC vision for exemplary science and mathematics education?
6. What is the extent of institutionalization of high quality professional development systems n the LSC districts?

LSC Core Evaluation Design

The LSC core evaluation used a mixed-methods design. Each project was required to have a local evaluator selected with the approval of NSF. The core evaluation required central reporting of evaluation data collected using pre-designed evaluation instruments to enable aggregation of data for national reporting.

The core evaluation instruments included questionnaires for teachers and principals, teacher interviews, and observations of both classroom and professional development sessions. Local projects could also add site-specific evaluation activities at their discretion.

HRI provided standardized protocols, surveys, and observation formats and the project-level evaluators were required to attend national sessions to learn how to use the instruments appropriately. HRI personnel initially assumed they would find existing instruments appropriate for the core evaluation. Finding none, they developed original observation protocols and surveys that would provide the quantitative results considered helpful in establishing the credibility of the observation data.

HRI convened a small advisory group of experienced people to discuss the data to be collected in the core evaluation. This group provided valuable insights into how the evaluation should be conducted as well as about the type of data to be collected. The group strongly recommended adding school-level tracking to the required data. NSF decided to make it optional, and interest in this area waned.[7]

LSC Core Evaluation Components[8]

As part of the core evaluation, local project evaluators collected data using standard questionnaires and protocols designed to answer the core evaluation questions. These evaluators were required to participate in meetings on data collection for the core evaluation. The meetings provided orientation and training on the use of the standardized data collection instruments provided by the core evaluation. After the initial meeting, refresher meetings were convened annually. In addition, HRI provided meeting attendees with training materials to facilitate the evaluators in providing training on the instruments and their use to others at the project-level. Finally, evaluators received special training on the classroom observation protocol through a mandatory two-day session that included observing and rating videotaped classroom lessons.

The local evaluators used several standard instruments in conducting the following activities:

Interviews with Teachers. Evaluators in each project interviewed a sample of classroom teachers who had participated in at least 20 hours of LSC professional development activities. The evaluators completed a form summarizing the interview data.

Observations of Classrooms. The core evaluation team randomly selected teachers for observation each spring at the beginning, middle, and end of the LSC project. These observations were guided by a standardized classroom observation protocol developed by HRI. The core evaluation analysis conducted by HRI included 1,620 observations.

Observations of Professional Development Activity. The core evaluation required evaluators to observe a number of professional development sessions annually using standardized protocols. Mature projects required 5-8 observations each year, while new projects were to conduct 2 observations to serve as a baseline. Overall, more than 2,400 observations of professional development sessions were conducted.

Project Ratings. PIs and evaluators jointly completed project ratings on the quality of their LSC professional development programs, on supportiveness of context, and likelihood of institutionalization.

Questionnaires for Teachers.[9] Each LSC project administered questionnaires to teachers, asking about their attitudes, beliefs, preparedness to teach, and instructional practices. Projects with 350 or fewer teachers administered questionnaires to all teachers; those with >350 teachers drew a simple random sample of at least 300. In all, nearly 75,000 teacher questionnaires were returned, including 36,828 from K-8 science teachers; 24,903 from K-8 mathematics teachers; 11,206 from 6-12 mathematics teachers; and 2,021 from 6-12 science teachers.

Questionnaires for Principals. Local projects administered questionnaires to principals in targeted schools. They were asked about their opinions about mathematics and science instruction, factors affecting mathematics/science instruction in their school, and their school's progress in implementing LSC reforms.

Questionnaires for PIs and Evaluators. Evaluators and PIs jointly completed project strategy questionnaires to provide information on the design and focus of the professional development program of each LSC.

Evaluation in Retrospect[10]

The HRI evaluation team concluded that collecting the same data from large numbers of participants over time enabled more sophisticated analyses than would otherwise have been possible. Many people at the project-level were trained in the use of the observation

instruments, thus increasing capacity in the field. The instruments were also adapted and used in other studies; some PIs used them as vehicles to create change, for example helping teachers understand the vision of standards-based instruction. Finally, some of the ideas from the core evaluation such as "activity for activity's sake" as a descriptor of poorly designed or implemented hands-on instruction have taken root as part of the conversation in STEM education.

The design of the core evaluation also led to a number of issues related to competing project- and program-level evaluation activities, as well as varying levels of experience across the locally-chosen project evaluators. For instance, some of the less experienced evaluators at the local level welcomed the prescribed instruments and technical support from the Core, while the more experienced evaluators saw the Core as more of an imposition which took attention and resources, leaving some local evaluation needs unmet.

The many lessons learned from the LSC core evaluation were sufficiently important to the NSF project officers, so that they requested that the core evaluation spend their 2-year no-cost extension period disseminating information, creating research briefs, submitting manuscripts, making presentations at professional meetings, and producing a book about expanding professional development throughout an entire district.

NSF Project Manager Perspective on the LSC Core Evaluation

This section of the case study examines one NSF program officer's perspectives on the agency's involvement in and use of the LSC core evaluation. Additionally, this section presents the program officer's beliefs about how use and involvement played out at the project level. We acknowledge that this data is based on the perspective of only one of several program officers. It is possible that other program officers may have stressed different points or ideas.

NSF Involvement. The NSF program officer reported being closely involved in formulating the direction of the LSC core evaluation especially during its planning stages. NSF contributed to the development of the evaluation's design, processes, and instruments through collaborative meetings and discussions with HRI and LSC project evaluators. In addition, NSF set the overarching agenda for the evaluation and worked on the details of data collection instruments, for example the program officer remembered "going through all those questions [on the classroom observation protocol] to see if we couldn't get a good idea… of what the science content was that was being delivered, how the teachers were learning, and whether it was manifesting itself in changed teaching in the classroom."

An important factor in explaining NSF's deep level of involvement in the core evaluation is related to how the evaluation was funded – the project was funded as a contract, as opposed to a grant. In a contracted project, the NSF acts as more of a client in requiring specific services and directing the form, content, and timing of contracted deliverables. By contrast, in a grant funded project, the PI would operate more autonomously in pursuit of the research goals

originally approved in the grant application. Another difference between grants and contracts is the level of compliance required with a range of regulatory provisions under the U.S. Office of Management and Budget (OMB), where the contract requirements are more numerous and complex than for grants.

Influence on NSF. The NSF program officer reported gaining a deeper understanding of evaluation in general; specifically how to better conduct program-level evaluation through experiences with the LSC core evaluation. As she reported,

> [W]hen you get involved in doing program evaluation like that, you begin to see things from quite a different perspective. The idea that this was a mechanism by which taking a sample from all the different projects we could… integrate all those separate pieces of information to come up with information about a Program. Now that may be obvious to lots of people, but to me it was a new experience.

The program officer also reported having learned that precise and appropriate instruments were not available to capture program objectives and that developing such instruments is a difficult task. She recalled when they learned that the tools and instruments they had expected to find did not exist but needed to be developed, saying,

> [T]he thing is, the more work you do the finer the grain needs to be on the instrument. …. [S]ome people in very powerful positions think the only thing needed is the funds. What they don't realize is the lack of instruments to measure what we need to measure to show that what we're doing is working.

Additionally, as a result of having NSF program officers attend evaluation planning meetings and because of their participation in developing data collection instruments, they became familiar with the projects and were better able to understand the project-level evaluation reports when they were submitted to NSF for their review.

NSF Perceptions of Project Involvement. The NSF program officer remarked that the projects were initially reluctant to get involved with the program-level evaluation. She explained that there was concern among project leaders and evaluators about participating in overarching program evaluation:

> …there was a lot of hesitancy on the part of evaluators to even sort of say, 'if we get involved in this and we tell you what's really happening, what's going to happen to us?' There was sort of a fear of the consequences of what would happen. I certainly had one evaluator express concern that if they were really honest they might get fired (laugh).

She believed that the idea of a program-level "core" evaluation was new to NSF grantees at the time, but today "people are used to the idea that programs are going to be evaluated" and therefore are less hesitant to participate in core evaluations.

The project officer reported that despite some concerns, the projects were deeply involved in developing the evaluation plan, processes, and instruments through participation in several planning meetings with the HRI, project PIs and staff, and the NSF program officers. Additionally, projects provided data to the core and the program officer saw evidence of the use of the data in the project-level evaluation reports that were submitted to NSF.

NSF Perceptions about the Influence on Projects. Overall, NSF felt that the involvement of project leaders and evaluators in the LSC evaluation contributed to a sense of ownership over the evaluation which led to a better, more useful evaluation. The program officer stated that she believes that "having people involved in creating something gives them more ownership of it," and because the projects felt ownership, they participated in an evaluation that otherwise they may have felt was imposed on them. As a result of their involvement, the NSF program officer felt that the core evaluation contributed to the development of a sense of community among the projects, increased the projects' evaluation capacity, built trust and understanding between NSF and the projects, and produced higher quality instruments and data. Additionally, the program officer was aware that LSC project staff and others used the instruments developed by the LSC core evaluation in other settings.

The NSF program officer noted that the LSC core evaluation developed a community in which there was a shared understanding of how to evaluate effective science and mathematics teaching. The program officer thought that the process of coming to a common understanding of science teaching and developing appropriate methods for evaluating science teaching "built evaluation capacity" of the projects through developing the system and requiring that people get trained to use it.

Additionally, developing the system collaboratively with NSF helped to establish trust between NSF and the projects. As the program officer shared,

> [T]rying to build rapport between NSF and its projects is very important because there's a tendency to only tell you the good stuff….If we could learn more about what doesn't work as well as what works, we'd be a whole lot better off because we wouldn't have to keep repeating the experiment. So trying to establish a level of trust between NSF and program evaluators is very important.

According to the program office, the timing of the evaluation, along with the involvement necessary to provide data, was important for understanding the evaluation's impact. She expressed a new appreciation of the importance of feedback loops and how the timing of the

evaluation and a particular project's funding could facilitate or limit that project's use of the data.

Project-Level Involvement and Use

This section of the case study examines the relationship between project involvement in the program evaluation and the use of the evaluation process and outcomes. It presents Project Leader and Evaluation Survey and interview findings organized according to the following topics:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

For each topic, quantitative results from the online survey will be discussed first. Data from project interviews will be addressed next. Relevant examples from the open-ended online survey items may also be included.

Motivation for Involvement

Project leaders (primarily PIs) became involved in the LSC core evaluation in a number of ways, and more of them reported volunteering or being invited than being required to participate. In fact, although PIs and evaluators were invited to participate in planning (both initially and as the instruments were refined and additional instrument developed); they were required to participate in data collection activities.

Table 4. Motivation for involvement – Comparing evaluators' and leaders' responses

Survey Question: Which option best describes the motivation for your involvement during the:	**Role**	**I was required**	**I was asked**	**I volunteered**	**N**
Evaluation planning stage *(chi-square = 18.345, df=2, p<0.001)*	Evaluators	15.8%	84.2%	0%	19
	Leaders	42.4%	24.2%	33.3%	33
Evaluation implementation stage *(chi-square = 8.359, df=2, p=0.015)*	Evaluators	39.3%	53.6%	7.1%	28
	Leaders	39.5%	26.3%	34.2%	38
Communication of evaluation findings stage *(chi-square = 6.223, df=2, p=0.045)*	Evaluators	42.9%	39.3%	17.9%	28
	Leaders	33.3%	20.5%	46.2%	39

In the evaluation planning stage, one third of LSC leaders responded that they volunteered to participate in the evaluation planning and implementation phases, while none of the evaluators reported having volunteered.

Overall, LSC project evaluators were less likely than project leaders to cite volunteering as their motivation for involvement in all three phases – planning, implementation, and communication.

This is not surprising given that although these evaluators were clearly not required to take the job, once they agreed to do so, there were definite evaluation activities that they were required to conduct. The NSF grant applications stated that participation in the LSC core evaluation was mandatory, particularly in the data collection elements.

More project leaders than evaluators reported being required to participate in planning and more evaluators reported being required to participate in communicating the findings. Given that the evaluators had to collect the data and share it with HRI and with the PIs by specific deadlines, these individuals might have classified entering data into web forms as communication rather than as data collection. In addition, evaluators might also have been more sensitive than others to the fact that evaluation planning had been completed without their involvement. For complete results see Table 4.

Among the interviewees, several factors emerged with regard to motivation for involvement including:

- Compliance with compulsory activity,
- Quality improvement,
- Accountability, and
- Relationships within a professional network.

Relative to compliance with compulsory activity, one project leader stated flatly, “I just sort of assumed that if you wanted your money every year, you had to fill out these things.”(LSC Interviewee 2) Another individual said, “Well, there was no choice. We were told we had to do this… the teacher survey was kind of viewed as something kind of annoying that we had to do.”(LSC Interviewee 3) And finally, the fourth interviewee said, “We just followed the rules.” (LSC Interviewee 1)

Another project leader focused on the idea of quality and performance improvement in addition to compliance, stating that the team was motivated to participate in the evaluation because “[T]here was this drive to do a good job…We just wanted the children and teachers to have the best possible experience.” In addition, accountability was also a motivating factor for this leader who noted feeling that “we had these evaluators standing over our shoulders – along with NSF.” (LSC Interviewee 1)

The interviewee went on to state that providing feedback to Congress for accountability purposes was an important factor for NSF, noting

> NSF gave a lot of federal government dollars, tax payer dollars, and I mean there has to be accountability. But it's really more than that, because they [also] want to find a way to raise a more scientifically literate society. (LSC Interviewee 1)

Finally, motivation for involvement in the evaluation planning phase was ascribed by one interviewee to an existing professional relationship that resulted in a personal invitation to participate. This interviewee noted knowing the LSC core evaluation PI for many years prior to this project. So when invited to participate in the project design and instrument development, the respondent accepted the invitation, because

> I have always found it enormously helpful to maintain a good network of colleagues. So, when people call or email me and ask me for help, I respond because it strengthens that network…And, I know I'm going to be on the receiving end of that give and take before too long. (LSC Interviewee 4)

Levels of Involvement

Evaluators reported significantly higher levels of involvement in the LSC core evaluation activities than did the project leaders. Evaluators' mean level of involvement in the 13 different activities included in the online survey was 3.0 compared to 2.5 for the project leaders. This difference was statistically significant ($t=3.165$, $df=71$, $p=0.002$). As a result, post-hoc tests were conducted on the 13 involvement items to see where the significant differences occurred. A Bonferroni adjustment was performed to control test-wise error rates; each test was compared to an alpha of .0038 (.05/13). The questions for which there was a significant difference between mean reported levels of involvement of evaluators and leaders are marked with an asterisk in Tables 5-7 below and the percentages are reported for each sub-group.

Involvement in Evaluation Planning Activities. Evaluators and project leaders reported similar levels of involvement in evaluation planning activities. As shown in Table 5, respondents reported highest levels of involvement in the discussions that focused the evaluation (mean=2.7). Approximately 44% of project leaders and evaluators indicated that they were extensively involved in the discussions that focused the evaluation.

Table 5. Involvement in evaluation planning activities

Survey Question: I was involved in…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
The discussions that focused the evaluation.	28.2%	15.5%	12.7%	43.7%	71	2.7	1.3
Developing the evaluation plan.	39.4%	14.1%	21.1%	25.4%	71	2.3	1.2
Identifying evaluation planning team members.	49.3%	8.5%	12.7%	29.6%	71	2.2	1.3

One possible explanation for the reported levels of involvement in focusing the evaluation is that occasional minor adjustments, in the process if not in the instruments, were made at the project meetings called by HRI.

Project evaluators and leaders reported lower levels of involvement in developing the evaluation plan (mean=2.3) or identifying evaluation planning team members (mean=2.2), although half or more of the respondents reported being involved at least a little in these activities. This result is consistent with the core evaluation structure in that the HRI evaluation plan and instruments were required, but projects could add to the evaluation at their discretion. In addition, NSF approval of the lead evaluator and others conducting observations for the core evaluation was required. About half of the respondents stated they were involved in identifying evaluation planning team members, and 60% reported that they were involved at least a little in developing the evaluation plan.

The interviewees differed in their reported levels of involvement in planning, with responses ranging from “no involvement” to “some involvement.” One PI said, “[I] do not believe I was involved in developing anything about it. But, we certainly participated and our teachers and principals filled out forms.” (LSC Interviewee 1) And another respondent said, “I did not work on the design of this [evaluation].” (LSC Interviewee 3)

By contrast, one interviewee reported having been invited to participate in the early planning stages with HRI based on pre-existing professional relationship stating, “[I had] some involvement at the very beginning, mostly in conceptualizing the design. But I wasn’t deeply involved.” (LSC Interviewee 4)

Table 6. Involvement in evaluation implementation activities

Survey Statement: I was involved in…	Role	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
Collecting data.* (p<.001)	Evaluators	0.0%	3.3%	13.3%	83.3%	30	3.8	0.5
	Leaders	25.6%	7.0%	37.2%	30.2%	43	2.7	1.2
Reviewing collected data for accuracy and/or completeness.* (p<.001)	Evaluators	3.4%	3.4%	17.2%	75.9%	29	3.7	.7
	Leaders	28.6%	21.4%	26.2%	23.8%	42	2.5	1.2
Interpreting collected data.* (p<.001)	Evaluators	0.0%	3.4%	24.1%	72.4%	29	3.7	0.5
	Leaders	33.3%	9.5%	38.1%	19.0%	42	2.4	1.2
Analyzing data.* (p<.001)	Evaluators	6.7%	3.3%	16.7%	73.3%	30	3.6	0.9
	Leaders	47.6%	19.0%	26.2%	7.1%	42	1.9	1.0
Developing data collection processes.	Both	39.7%	19.2%	13.7%	27.4%	73	2.3	1.3
Developing data collection instruments.	Both	41.7%	20.8%	25.0%	12.5%	72	2.1	1.1

Involvement in Evaluation Implementation Activities. Table 6 shows that evaluators and project leaders reported similar levels of involvement in developing data collection processes and instruments. In both cases, approximately 60% of respondents reported being involved at least a little in those activities. Evaluators and leaders reported different levels of involvement in the other evaluation implementation activities and evaluators consistently reported higher levels of involvement than did the project leaders.

All of the evaluators responding to the survey reported being involved in data collection and interpretation at least a little and over 90% reported being involved in reviewing and analyzing data. By contrast, about three quarters of project leaders reported involvement in data collection and reviewing data, about half reported involvement in data analysis, and two thirds reported involvement in interpreting the data.

The projects were required to write annual project reports using data from the core evaluation, along with rating their own project's performance on the evaluation questions. Project evaluators collected the evaluation information, including their ratings of what they observed, and then submitted it to HRI. For example, for an evaluative interview with a teacher, the evaluator would rate the teacher's responses and provide HRI with a quotation in support of that rating. These activities are consistent with the reporting, analyzing, and/or interpreting categories included in the survey.

One interviewee (an evaluator) noted having had involvement in implementation that included:

> …distributing and collecting all the questionnaire data every year from teachers and principals…and there were interviews associated with that. …When I wrote the report for [my project] and for NSF – including the one I wrote based on my own [evaluation] design – I incorporated the HRI data. So, in other words, I had to work with the analysis of the data from the questionnaires. (LSC Interviewee 3)

As in the previous section, two of the interviewees described having had some involvement in the implementation of the core evaluation insofar as they collected data or provided data as a survey respondent. One interviewee commented,

> I had a private coordinator and really that was her job to get all the stuff to the principals and to the teachers and so forth and make sure it got collected….I was personally involved [in that] when they asked the PI to fill a form out, I always did that one. (LSC Interviewee 2)

Another interviewee reported having had more extensive involvement in the implementation phase of the LSC core evaluation, saying

> We actually contracted with Horizon to do additional data collection in our district so we could have district-level data. We wanted district-wide data and…to be candid, because of what I saw as a flaw in their strategy, [HRI's] sampling wasn't going to give us that. We also didn't want to administer two surveys, so we used our money and went to HRI and had them survey all of our teachers. We had them go beyond their sample because we needed district-wide data, and their sample wasn't designed to do that. (LSC Interviewee 4)

Finally, in response to a question about the level of involvement, one interviewee equated involvement with participation in meetings:

> There would occasionally be PI meetings. There were two or three of them I went to. And Iris Weiss would speak those meetings. And you have to realize this feels like a lifetime ago. We'd have breakout groups and things like that. And Iris would present … summaries of how the core evaluation was going and maybe ask for input. (LSC Interviewee 2)

Involvement in Evaluation Communication Activities. Consistent with the findings related to data collection and related activities discussed above, evaluators and project leaders reported different levels of involvement in writing evaluation reports. Over 83% of evaluators reported being extensively involved in writing evaluation reports**.** As shown in Table 7, the mean involvement of evaluators was 3.7 out of 4 compared to only 2.0 for project leaders. Evaluators and leaders reported similar levels for the remaining evaluation communication activities, i.e.,

reviewing reports for completeness, presenting evaluation findings, and developing future project plans. In all three of these activities, approximately half of the respondents reported being extensively involved in the activities and around one in six were not involved at all in the activity.

Table 7. Involvement in evaluation communication activities

Survey Statement: I was involved in…	**Role**	**No**	**Yes, a little**	**Yes, some**	**Yes, extensivel**	**N**	**Mean**	**SD**
Writing evaluation reports.* (p<.001)	Evaluators	3.3%	3.3%	10.0%	83.3%	30	3.7	0.7
	Leaders	46.3%	19.5%	24.4%	9.8%	41	2.0	1.1
Presenting evaluation findings.	Both	16.7%	8.3%	26.4%	48.6%	72	3.1	1.1
Developing future project plans based on evaluation results.	Both	18.3%	8.5%	22.5%	50.7%	71	3.1	1.2
Reviewing evaluation reports for accuracy and/or completeness.	Both	16.9%	12.7%	19.7%	50.7%	71	3.0	1.2

Perceived Effect of Involvement

Overall, LSC evaluators and project leaders responded that the input they provided to the core evaluation was considered and used. As shown in Table 8, over 95% reported that their input was considered at least a little and over 90% reported it was used at least a little during the planning, implementation, or communication stages.

Interviewees reported that they were unsure whether or how their input to the core evaluation might have been used. When asked whether input offered had been used, one interviewee said, “I have no idea.” (LSC Interviewee 3). Another remarked, “Well, I don’t really know that she used any of my input….So, I have no idea how influential my involvement was.” (LSC Interviewee 4)

Table 8. Consideration and use of project leader and evaluator input

Survey Statement: The input I provided during the…	No	Yes, a little	Yes, some	Yes, extensively	N*
Evaluation planning stage was taken into consideration.	4.4%	24.4%	33.3%	37.8%	45
Evaluation implementation stage was taken into consideration.	3.2%	22.6%	32.3%	41.9%	62
Communication of evaluation findings stage was taken into consideration.	1.7%	16.9%	32.2%	49.2%	59
Evaluation planning stage was used.	9.3%	16.3%	39.5%	34.9%	43
Evaluation implementation stage was used.	5.3%	15.8%	43.9%	35.1%	57
Communication of evaluation findings stage was used.	3.6%	12.5%	41.1%	42.9%	56

*Total N is small because some individuals responded that they did not provide input or they did not know if their input was considered or used.

Another interviewee noted,

> They [HRI] did those little booklets that they published as a cumulative of the past year trying to look across all the LSCs after the fact. Some used quotes from our annual reports, so they had used some of our data. So, in that sense I know they used some of our things. (LSC Interviewee 2)

From a third to half of survey respondents reported that their input was extensively considered and used across the different evaluation phases included in the survey. Possible reasons for these responses are that respondents may feel that the input they provided during the meetings sponsored by the core evaluation was listened to and taken into account by HRI staff in trying to improve the process and be responsive to the project PIs and evaluators. Some respondents might also have been thinking of the use of their evaluation findings in the annual high-level summaries compiled by the core evaluation team.

Perceived Use by Project Leaders and Evaluators

The online survey asked project leaders and evaluators to reflect on the extent to which the LSC core evaluation influenced their knowledge and understandings, skills, and beliefs about evaluation. Additionally, project leaders were asked about their use of the core evaluation process and findings to make decisions about their project or in other projects.

As shown in Table 9, data from the survey suggest that the LSC core evaluation had the greatest influence on project leaders' and evaluators' knowledge/understanding (mean=2.9) and skills

(mean=2.9), followed closely by influencing their beliefs (mean=2.7) and use of the evaluation in other contexts (mean=2.7). There were no significant differences found between project leaders and evaluators in the use or influence of the evaluation for any of these questions.

Table 9. Influence of core evaluation on project leaders and evaluators

	N	Mean	SD
Influence on knowledge/understanding (q25-30)	72	2.9	0.8
Influence on skills (q31-36)	72	2.9	0.8
Influence on beliefs (q38-43)	72	2.7	0.9
Use of knowledge, products, processes in other contexts (q45-49)*	57	2.7	0.9
Ways of using findings (q50-58)**	44	2.6	0.8

*items asked only of those reporting having conducted a subsequent evaluation
**items asked only of project PIs

Table 10 displays reports of project leaders and evaluators relative to the influence of the core evaluation on their knowledge and understanding. Overall, project leaders and evaluators reported that the core evaluation had the greatest influence on their knowledge and understanding of their own projects (mean=3.2), followed by their understanding of how to implement an evaluation (mean=3.0), how to plan an evaluation (mean=2.9), and how to communicate evaluation findings (mean=2.8). The project leaders and evaluators also responded there was some influence on their understanding of STEM education evaluation (mean=2.8), however this did not translate into as much of a gain in understanding of STEM education generally (mean=2.5).

Table 10. Influence of the LSC core evaluation on project leaders' and evaluators' knowledge and understanding

Survey Question: The evaluation increased my knowledge/understanding of…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
My project.	5.6%	13.9%	37.5%	43.1%	72	3.2	0.9
How to implement an evaluation.	5.6%	23.9%	31.0%	39.4%	71	3.0	0.9
How to plan an evaluation.	11.1%	20.8%	37.5%	30.6%	72	2.9	1.0
STEM education evaluation.	9.9%	18.3%	46.5%	25.4%	71	2.9	0.9
How to communicate evaluation findings.	11.1%	23.6%	36.1%	29.2%	72	2.8	1.0
Science, Technology, Engineering, and Mathematics (STEM) education.	15.7%	37.1%	31.4%	15.7%	70	2.5	0.9

Influence on Knowledge and Understanding. The interviewees identified several instances where the core evaluation influenced their knowledge and understanding, particularly in the areas of STEM education and the evaluation of STEM education.

One interviewee described the importance of the information generated from the classroom observation protocols, stating, "The classroom observation protocol was probably the best thing, in my opinion, that came out of that whole core evaluation." (LSC Interviewee 2)

In addition to the classroom observation protocol, this interviewee singled out the new knowledge generated by the LSC core evaluation relative to the number of professional development hours it takes to make a difference. The interviewee said, "It was informative from the perspective of how many hours of professional development teachers really need to have before they can start to make an impact." (LSC Interviewee 2)

Influence on Skills. LSC project leaders and evaluators reported that the evaluation had the greatest influence on their skills for working on their project (mean=3.1), followed closely by their skills in planning, implementing, and communicating about their evaluation and STEM education evaluation.

Table 11. Influence of the LSC core evaluation on project leaders' and evaluators' skills

Survey Question: The evaluation improved my skills…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
For working on my project.	5.6%	18.1%	38.9%	37.5%	72	3.1	0.9
In implementing an evaluation.	6.9%	20.8%	36.1%	36.1%	72	3.0	0.9
In communicating evaluation findings.	11.3%	14.1%	38.0%	36.6%	71	3.0	1.0
As a STEM education evaluator.	12.7%	15.5%	38.0%	33.8%	71	2.9	1.0
In planning an evaluation.	11.1%	23.6%	31.9%	33.3%	72	2.9	1.0
As a STEM educator.	22.5%	26.8%	36.6%	14.1%	71	2.4	1.0

The interviewees identified the training associated with the classroom observation protocols as being responsible for new skill acquisition. As one interviewee explained,

> I would say the impact on me came mostly from the training in the use of the classroom observation protocol. The training and the discussions within the groups at the training….this was particularly useful to me….But I can't say that the LSC project core evaluation directly changed any of the evaluation skills that I have. (LSC Interviewee 4)

Another person noted that the impact of the training associated with the classroom observation protocol, was clear because "…we had to be certified. [And] we had to observe some videos and score them, to produce some inter-rater reliability." (LSC Interviewee 3)

Table 12. Influence of the LSC core evaluation on project leaders' and evaluators' beliefs

Survey Question: The evaluation increased my belief in the importance of…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
Planning an evaluation.	13.9%	19.4%	26.4%	40.3%	72	2.9	1.1
Implementing an evaluation.	16.9%	14.1%	29.6%	39.4%	71	2.9	1.1
Communicating evaluation findings.	15.3%	13.9%	34.7%	36.1%	72	2.9	1.1
My project.	23.9%	8.5%	38.0%	29.6%	71	2.7	1.1
STEM education evaluation.	27.1%	14.3%	30.0%	28.6%	70	2.6	1.2
STEM education.	39.4%	15.5%	29.6%	15.5%	71	2.2	1.1

Influence on Beliefs. As shown in Table 12, fewer survey respondents reported that the core evaluation influenced their beliefs than their knowledge and skills (as displayed in Tables 10 and 11). Additionally, the evaluation did not appear to influence their beliefs about their projects (mean=2.7) to the same extent it did their knowledge and skills related to their project, where both means were greater than 3.0. However, reported influence on beliefs related to the three stages of conducting an evaluation was greater than reported influence on project leaders' and evaluators' beliefs about STEM education in general (mean=2.2).

One interviewee described the importance of the paradigm-shifting insight relative to the change in thinking and beliefs about the value of activity in the education field. This interviewee noted,

> [W]e had the mentality that any inquiry or any hands-on activity was good. And, so it was a nice way to start to think about that not all inquiry was good. It had to be high-quality inquiry, not just 'activity for activity's sake.' So, in the sense of looking at inquiry as hands-on activity, again, I think that [classroom observation] protocol impacted me personally. (LSC Interviewee 2)

The interviewee continued,

> So, these NSF meetings or Horizon meetings, I guess, coincided with that whole national conversation….An artifact of being a part of a large national project at the time this is all going on is that it got many of us who were involved as PIs thinking along these lines. (LSC Interviewee 2)

Influence on Future Behavior. Among those respondents who have conducted an evaluation since the LSC evaluation, about 90% of respondents reported that they used what they learned at least a little in another context; approximately 70% of reported having used the LSC program evaluation plan as a model for another evaluation; and 87% reported using the LSC data collection instruments in another evaluation context. See Table 13.

One of the interviewees described having used the classroom observation protocols in other settings,

> We used [them] quite extensively in the training of our lead teachers and in the lead teachers helping their peers. So, even though it was supposed to be – from the core evaluation's perspective – part of that core data collection, we ended up using it as a way of helping teachers understand what a good lesson looked like. (LSC Interviewee 2)

Table 13. Use by project leaders of knowledge, products, processes in other contexts

Survey Statement:	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
I used what I learned from planning the LSC program evaluation in another evaluation.	9.3%	13.0%	37.0%	40.7%	54	3.1	1.0
I used what I learned from implementing the LSC program evaluation in another evaluation.	9.3%	20.4%	38.9%	31.5%	54	2.9	0.9
I used data collection instruments from the LSC program evaluation in another evaluation.	13.0%	16.7%	50.0%	20.4%	54	2.8	0.9
I used what I learned from communicating evaluation findings from the LSC program evaluation in another evaluation.	22.2%	18.5%	29.6%	29.6%	54	2.7	1.1
I used the evaluation plan from the LSC program evaluation as a model in another evaluation.	29.6%	22.2%	31.5%	16.7%	54	2.4	1.1

Another interviewee stated that the LSC core evaluation helped "drive future work," adding,

> We actually created a shorter version of it which we gave to all of our principals. And we created a different version – we improved upon it and moved away from the forced judgment at the end – which we are using right now [in another NSF project]. (LSC Interviewee 4)

One of the interviewees described using the core survey instruments in other NSF evaluations for pragmatic, if not scientific reasons, saying,

> We know that the NSF likes the surveys that were created. And so…we decided to use the HRI surveys for [another project.] We didn't do that because we particularly liked that survey or particularly believe in it, but because we felt that we'd collect some pre-post data that we could present to the National Science Foundation. (LSC Interviewee 3)

Use by Projects

In addition to asking questions about the individual impact of the LSC core evaluation (addressed above), project PIs were asked questions about ways in which they used the LSC program evaluation findings within their projects. As shown in Table 14, PIs reported moderate levels of use of the findings to meet contractual requirements for their projects (mean=3.0), develop future plans for their projects (mean=2.8), and make changes to their project (mean=2.8).

Approximately 85% of project PIs indicated they used the program evaluation findings at least a little to support decision making and grant requirement activities. In the case of using the findings to meet grant requirements, 48% of PI respondents reported that they used the findings extensively.

One interviewee noted a contrasting opinion stating, “I don't think the findings were used to make decisions because they weren't particularly useful for this project.” (LSC Interviewee 3)

The other interviewees, however, identified examples of use of the findings in talking to community leaders and potential funders. One appreciated having the findings in situations where recommendations were needed. The interviewee said,

> Frequently, in talking to funders or to high level leaders or to the community, they’ll say, ‘Well maybe you should be doing this or that.’ But you can come back and say, ‘This is what the research shows.’ And so I think that research and referring to it helps the vision. (LSC Interviewee 1)

Another interviewee used the survey data “in reporting to our Advisory Council,” noting,

> I was not very fond of the survey itself because it had too many god, motherhood, and apple pie questions – questions with clearly expected right answers, which, in my opinion, ended up exaggerating the positive aspects of implementation to an extent. So, in our case, when we were sitting down with our Advisory Council, we had to explain the discrepancies. Because there would be survey data showing that everyone reports doing these good things, but the observation data showing no they weren’t really doing any of these good things. (LSC Interviewee 4)

Table 14. Ways project PIs used LSC core evaluation findings

Survey Statement: I used the LSC program evaluation findings to…	No	Yes, a little	Yes, some	Yes, extensively	N*	Mean	SD
Meet contractual and/or legal requirements of my LSC project.	15.9%	15.9%	20.5%	47.7%	44	3.0	1.1
Develop future plans for my LSC project.	15.9%	15.9%	36.4%	31.8%	44	2.8	1.1
Make changes to my project(s).	11.4%	22.7%	40.9%	25.0%	44	2.8	1.0
Help plan a new project(s).	25.0%	9.1%	38.6%	27.3%	44	2.7	1.1
Enhance my organization's commitment to my LSC project.	31.8%	15.9%	15.9%	36.4%	44	2.6	1.3
Make decisions about the future existence of my LSC project (e.g., cancel or continue).	25.0%	22.7%	27.3%	25.0%	44	2.5	1.1
Increase the attention given to my LSC project.	34.1%	15.9%	22.7%	27.3%	44	2.4	1.2
Seek funding for my project(s).	43.2%	13.6%	18.2%	25.0%	44	2.3	1.3
Promote additional evaluation of my project(s).	40.9%	25.0%	18.2%	15.9%	44	2.1	1.1

*Only project PIs were asked this question, hence the lower N.

Factors Affecting Use

The interviewees identified one general category of factors that they believed might have enhanced the overall level of influence of the LSC core evaluation, and three categories identified as barriers to influence.

Enabling Factor. All of the interviewees identified the quality and rigor of the instruments and overall evaluation process as a positive factor in lending credibility and thus influence to the core evaluation products and activities.

For example, one interviewee characterized the core evaluation as being "important with the powers that be." (LSC Interviewee 1) And, another interviewee noted that "they gave us instrumentation and classroom observation protocols that were pretty well done." (LSC Interviewee 4)

The interviewees were more talkative in response to a question about what factors could be identified as barriers or inhibitors to the potential to influence of the core evaluation. The barriers identified fall into three areas: (1) timing, (2) lack of alignment, and (3) information overload.

Challenging Factors. Interviewees commented that the timing of the core evaluation created more work and added costs to the local projects. Specifically, one interviewee noted that "no time was allocated for the national evaluation because it didn't exist at the time [the local evaluation was developed.]" This interviewee added that the core evaluation was "time put on top of what I needed to do. And I resented that….because it took a good amount time…and impinged on the evaluation activities that I could do insofar as none of us as evaluators want to create burdens on teachers." (LSC Interviewee 3)

On the subject of funding, another interviewee noted,

> I can't speak to the funding because I don't know anything about [the core evaluation's] budget. But it was an awkward situation because this whole thing was layered on to what we had already designed and put in place. …it came after project evaluations were already in place. It was just layered on top. (LSC Interviewee 4)

This same interviewee commented on another aspect of timing relative to the local LSC project saying, "NSF had already abandoned LSC and was moving on to the new and improved MSP." This interviewee added that "[T]he news of LSC wasn't all positive. And about the time the report was coming out … I don't think [NSF] wanted to hear any bad news."(LSC Interviewee 4)

One interviewee pointed out the perceived lack of alignment between the core's evaluation activities and the activities, needs, and challenges of local projects. This interviewee noted that this lack of alignment…

> …wasn't taken into account by the national survey…. [Every year] I had to write a report explaining the ways in which the national evaluation was not aligned with this project…. [T]here are questions in the survey asking teachers about their involvement in LSC activities. And those teachers did not know what activities they were doing that were LSC activities. So, that data was not accurate…(LSC Interviewee 3)

Several interviewees noted their impressions of information overload caused by the core evaluation. They described a sense of frustration at their inability to keep up with reading and summaries that were issued by HRI. An interviewee described it as follows,

> I wish that there had been more summaries made or sort of thrown in our face….I wish it had been easier to access the summaries because I didn't have time to sit down and read everything they put out. (LSC Interviewee 1)

High Users' Perspectives

The following is a presentation of data from interviews with three survey respondents designated by the research team as high users based on their mean use/influence score on the

online survey. Interviewees were selected based on mean use/influence score only, without consideration of involvement levels or roles. These individuals had all consented to be contacted after the survey for follow-up interviews, and they had not been previously interviewed by the by the project team. The three high user interviewees were all project evaluators; no project leaders were included in the high user interview sample. The presentation of the high user interview data is organized according to the same six categories as above:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

Motivation for Involvement. Similar to the other interviewees, the interviewees in the high use group also identified compliance with required activity and relationships within a professional network as reasons for their involvement in core evaluation activities.

Two interviewees explicitly identified connections within their personal networks that led to them becoming involved as external evaluators on their respective LSC projects. In one case, the individuals' dissertation committee chair was a project leader and requested the evaluator's involvement. In another case, the interviewee had been a research collaborator with the project PI and they shared an interest in professional development and capacity building. So the project PI asked the interviewee to join the project as the external evaluator.

Levels of Involvement. Each of the high-use interviewees reported that they were involved to some degree in the core evaluation. These interviewees all reported having had no involvement in planning the core evaluation. This finding is consistent with the core evaluation structure wherein the HRI-designed instruments had to be used by all projects. In addition, none of the high-use interviewees was involved at the start of the core evaluation when they might have been asked for feedback as HRI developed the evaluation design.

All of the high-use interviewees reported that they were involved in three specific activities – a) training, b) workshop-attendance, and c) data collection activities as part of the implementation phase of the core evaluation. Two of the three pointed to their involvement in data analysis and report-writing, with one interviewee identifying that these two activities took place in the form of having to write an "analytic memo…in a format that Horizon gave us." (LSC Interviewee 6)

Perceived Effect of Involvement. While none of the high-use interviewees could point out specific examples of exactly where and how their input had been used by the core evaluation

team, they all agreed that their input was solicited and considered by HRI. One interviewee stated

> "I think they took it into consideration…I think Horizon was sensitive to massaging and adjusting the classroom observation protocol to reflect more of a constructive approach – an open-ended approach to science teaching." (LSC Interviewee 5)

Another interviewee commented on feeling listened to by the core evaluation. This interviewee commented on adjustments to the data collection instruments that were discussed at the final meeting on the 10th anniversary. He said,

> "I don't know if it was my opinion that was taken into consideration or everybody's collective opinion….[but] I felt a little bit of ownership because those were some of the comments I had offered." (LSC Interviewee 7)

Use by Project Leaders and Evaluators. The interviewees varied somewhat in their assessments of the level of impact the core evaluation had on each them of personally: two of the three said it had some impact on them, and the third said it had extensive impact.

All three said that they used the instruments from the core evaluation or variations thereof in other settings. One interviewee also noted using other reports and information from HRI that he became aware of as a result of the core evaluation.

All of the high-use interviewees said they learned new evaluation skills as a result of the core evaluation. One person cited having acquired new skills, specifically, "interview techniques, ways of documenting interviews, ways to…make similar judgments and use some of our coding across interviews…." (LSC Interviewee 6)

This same interviewee also noted that the core evaluation changed how he felt about the use of evaluation saying,

> "Basically I think it made us look more systematically. Had I just gone into classes and started observing about what's going on here, my observations would have been much more vague and less specific and less useful that way. I think in the fact of knowing that the data I was collecting was being done in way to be compiled with other people's data… gave us a sense of the importance of what we were doing is serving on a national level and helped us get past the frustrations of not succeeding as well on the local level…" (LSC Interviewee 6)

Another interviewee reported that the LSC core evaluation had an extensive impact on him. He said his feelings about the importance of evaluation changed,

> "… in the sense that you need to have some kind of evidence to show anybody whether something is working or not. And it's not enough for the project to know what's going but I think that the world should know, too." (LSC Interviewee 7)

He closed by saying, "Personally I think I have learned a lot about evaluation and that's about all I can say." (LSC Interviewee 7)

Use by Projects. The high-use interviewees agreed that the core evaluation had an impact on their projects, specifying that information gathered at core evaluation workshops was used to adjust their interpretations when using the evaluation instruments (e.g., classroom observation protocol) to be more in line with the interpretations described by evaluators in other projects.

One interviewee mentioned using the core evaluation meetings for "calibration" and for looking at "how we stand as compared to the other groups." This interviewee went on,

> "When I realized other people were interpreting a given question on the observation protocol, for example, in a different way than I was, I'd say OK, I'll try to do it more in accord with what they are interpreting." (LSC Interviewee 6)

All of the interviewees also agreed that the core evaluation resulted in changes at the local level. For instance, one interviewee said, "It forced us to change our agenda." (LSC Interviewee 5) Another said, "Absolutely. We did a lot of changing." (LSC Interviewee 6)

Two of the three high-use interviewees specified that they used the core evaluation to make decisions about their local projects. One interviewee illustrated that statement as follows:

> "The data that I collected about our specific project I know was used both as a formative evaluation and as a summative evaluation. So information on the professional development activities or information from the teacher interviews or classroom observations were the main ones that probably influenced the adjustment in our approach to doing the project and also was confirmatory or supportive of some decisions that were made along the way." (LSC Interviewee 5)

Finally, one of the high-use interviewees identified that the process of going through the training and data collection affected the way that the project staff interacted by saying,

> "In the initial two years if I said something [was] going to impact the teachers, they were hesitant to use it. But through the program evaluation, we took some of the teachers to PI meetings. I think it definitely made a difference and they had better communication to be able to address some of these items." (LSC Interviewee 7)

Factors Affecting Use. The high-use interviewees identified the training meetings as the primary enabling factor, and burdensome requirements, timing, and lack of resources as the three factors that served as barriers to use of the LSC core evaluation in general.

Enabling Factors. One of the interviewees noted that the training meetings were a very big factor in facilitating use of the LSC core evaluation saying,

> "Certainly the training meetings that we had over those years enhanced and helped a lot. They were definitely very useful in many, many ways. They were really well done and really helped. Comparing notes with evaluators across the country. Comparing notes from the Horizon people themselves about what it was they were looking for and what they had learned. Just that collegial exchange of information and ideas…" (LSC Interviewee 6)

In addition, this interviewee went on to state that

> "…the fact of having to do the evaluation made [us] more thoughtful about what was happening and able to communicate what was happening and what things needed changed and could be improved." (LSC Interviewee 6)

Challenging Factors. One of the high-use interviewees reported being unable to identify any factors that served as barriers to the use of the LSC core evaluation, while two high-use interviewees identified three areas that served as barriers: timing, burdensome requirements, and lack of resources.

First, consistent with other interviewees, one high-use interviewee identified timing as a barrier to increased use of the LSC core evaluation. Specifically, this interviewee noted that being involved more at the very beginning would have contributed to greater use of the evaluation. The interviewee said,

> "[I]f our project had been funded that first year along with Horizon's contract, those projects probably had more influence on what Horizon did because they were founding members of the LSC network and by the time we came in, a lot of those things were in place and they couldn't be

> changed because baseline data had already been collected. I think that would have helped us." (LSC Interviewee 5)

Next, the requirements of the LSC core evaluation were identified by two high-use interviewees as a limiting factor relative to the use of the LSC core evaluation. Describing the problems caused by "burdensome requirements" one interviewee commented,

> Do you need questionnaires as regularly as questionnaires seem to be administered...I think there were probably other parsimonious procedures and decisions that could have been made to save time. Some of our teachers seemed to get…hit every year and they became somewhat concerned and especially in terms of the classroom observations. They thought we were trying to track their changes … and we kept saying to them, no, this is a random sample and you just happened to get caught twice in the three samples.

This interviewee closed by saying, "[T]his concept of social capital is crucial….There is only so much time and energy that you have in a project and it has to be used very strategically. (LSC Interviewee 5)

Another high-use interviewee noted,

> [Spending] a lot of time trying to collect data showing that the professional development really made a difference in the schools to the students in terms of matching up with the grades on standardized tests… turned out to be practically totally impossible because of turnover in students and turnover in faculty…. In that sense that piece was an exercise in frustration and a huge waste of time. (LSC Interviewee 6)

Finally, one of the high-use interviewees listed lack of resources as a problem reporting,

> [We] didn't set aside enough resources and finance to carry on the tremendous costs of putting on some of these things. [For] classroom observations we way underestimated the cost of that because of our project involving two states…there were travel costs and personnel costs that we didn't consider. That was just idiosyncratic to our naivety and lack of awareness." (LSC Interviewee 5)

Project-Level Summary

This case presents evidence of the impact of the core evaluation on individual project leaders' and evaluators' knowledge, skills, and beliefs, as well as on their future project activities.

The case findings are consistent with the notion that involvement in the evaluation activities is associated with the use of the evaluation in later projects and activities. In this case, evaluators reported significantly higher levels of involvement in the LSC core evaluation activities than did the project leaders. With respect to the motivation for that involvement, more people reported having volunteered to be involved than required, despite the fact that participation in the LSC core evaluation was mandatory, particularly in data collection activities.

According to the individuals interviewed and surveyed, a majority reported that their input to the evaluation process was considered and used by the core evaluation team, despite the fact that the core evaluation was designed by a small advisory team in conjunction with the core team at HRI. However, this might also be a testament to the opportunities for exchange (e.g., PI and evaluator meetings) provided to the projects as part of the core evaluation activities. It should also be noted, however, that the interviewees were uniformly unsure whether or how their input to the core might have been used.

The individuals interviewed were generally in agreement about the influence and impact of the LSC core evaluation on themselves, on their projects, and on the field. In terms of impact on decision-making, among the individuals surveyed, 85% or more of respondents indicated they used the program evaluation findings at least a little to support their decision-making and contractual requirement activities.

Finally, the interviewees identified quality as a facilitating factor, and three categories were identified as barriers to influence, including: (1) timing, (2) lack of alignment between the LSC core evaluation and project-level evaluations, and (3) information overload. The high-use interviewees also identified timing as a barrier, and they added two additional limiting factors: (1) burdensome required activities and (2) lack of resources.

Influence on the Field

This section of the case study examines factors that appear to be related to the use and influence of the LSC core evaluation on the fields of STEM education and evaluation. It presents findings resulting from four different data collection activities and is organized according to the following three topics:

- Projects' Perceptions of Influence on the Field
- The "Field's" Perception of Influence on the Field
- Evaluation Products' Influence on the Field

First, all survey respondents and interviewees were asked directly about their perceptions of the impact of the core evaluation on the field. Second, to assess the influence of these evaluations on the broader fields of STEM education and evaluation, the research team used two data collection methods: a survey of NSF PIs and a brief survey of relevant journal editors. The NSF PI survey was designed to assess whether the LSC evaluation, its PIs, and related publications would appear among the names and titles of influential evaluation-related people, products, and activities. The brief email/telephone survey of journal editors of six journals in the evaluation and STEM education fields was conducted to determine editor awareness of influential STEM evaluations. Finally, as another measure of influence on the field, citation analysis was conducted to examine the extent to which the LSC core evaluation products were cited in scholarly articles in the field.

Projects' Perceptions of Influence on the Field

Our first set of findings related to influence on the field was drawn the Project Leader and Evaluator Survey and interviews. Respondents were asked directly about their perceptions of the impact of the core evaluation on STEM education and evaluation. The responses from the survey and all interviews are summarized below.

As shown in Table 15, 93% of survey respondents reported that the LSC program evaluation had influenced the STEM community to some degree (15% said it had been very influential) and 96% of survey respondents reported that it had influenced the evaluation community to some degree.

One interviewee associated with a peer review journal in the field offered evidence of this influence as follows,

> I see a lot of manuscripts that come through [where] the LSC classroom observation protocol has been used in so many grants and so many other programs nationwide that I think it really was never intended to be that way…. they were intended for the LSC, but subsequently so many other programs around the country have used that observation protocol. (LSC Interviewee 1)

By contrast, another stated, "I don't remember that it had any impact whatsoever." (LSC Interviewee 3)

Table 15. Influence of program evaluation on the field

Role	Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
STEM Community	5	19	21	15	11	71
	7.0%	26.8%	29.6%	21.1%	15.5%	100%
Evaluation Community	3	16	31	13	6	69
	4.3%	23.2%	45.0%	18.8%	8.7%	100%

And a third interview commented on the impact in the context of timing stating,

> I'm not sure it had that much impact when we were doing the project. It's almost more after we were finished. That retrospective piece has impacted some of the professional development activities we're running at our university and some of the new grants we've written. So, I guess – to use the phrase that NSF uses in grants: it's more like lessons learned that helped us from that point on. (LSC Interviewee 1)

Among the high-use interviewees, two of the three identified an example of the impact of the core evaluation on the field of STEM education.

One interviewee said

> I think it clarified that …the series of one shot workshops and what I call "butt time" … simply exposure is not necessarily a good variable to use to assess the effects of professional development with teachers. But at the same time… doing 130 hours, which seemed to be an arbitrary number, the last 30 hours may not have the same impact as the middle 40 did. That's coming out of the data analysis that Horizon is doing. (LSC Interviewee 5)

Another high-use interviewee noted:

> "I think there is more awareness of evaluation…and I think there have been other projects that I know of that have also used the same kind of instruments. I think people are more aware of the importance of doing evaluation." (LSC Interviewee 7)

In general, the three high-use interviewees agreed that it was difficult for them to comment on the impact of the core evaluation on the impact on the evaluation field more broadly. Two of them felt that it is probably not realistic to expect these kinds of broader impacts from large-scale program evaluations.

In terms of the lasting impact of the core evaluation, the interviewees did not think that had been determined yet. One interviewee noted,

> I think that's yet to be determined. I have tried to within the Science Ed community to get people to document the lessons learned in both program evaluation and in professional development for science teachers and science programs at the system level. I have not yet been able to get a group interested in doing that as a book, for instance. . . They may be so tired from completing the project that they just want to see the back side of, too. Maybe in 10 years. (LSC Interviewee 5)

Another interviewee commented

> The problem is you'd have to have another evaluation to find out about the use of the evaluation. Show how you'd have to figure out how to track who is doing the reading, who read the reports and so and so. I don't know. There may have been a broader effect of this evaluation. (LSC Interviewee 6)

The "Field's" Perception of Influence on the Field

The research team attempted to measure the field's perception by surveying NSF PIs and journal editors. The first of these two was the NSF PI survey which gathered PIs' perceptions related to evaluations, people and publications. The researchers reasoned that if respondents mentioned the names and titles of the LSC program evaluation, then this might indicate the presence of a network of influence among the four targeted programs and STEM educators and evaluators.

The research team examined the survey results for evidence of influence of the LSC evaluation, its PI, and related publications on the broader fields of STEM education and evaluation. Sixty-four of the 191 respondents indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. The LSC evaluation was named by four of the 51 respondents. Similarly, of the 44 respondents who named a publication, four named an LSC publication. Finally, in response to the question regarding influential STEM evaluators, seven of 59 respondents named an evaluator associated with LSC – the greatest single response to this question.

Focusing further, there were two respondents who reported that they were involved in LSC, one respondent identified as influential an LSC-affiliated evaluation, publication, or evaluator by name.

As a second look at the field's perception of influence on the field, the research team conducted an 3-question survey of STEM education and evaluation journal editors. One of the three editors that responded to the journal editor survey named one STEM education evaluator who was the PI of the LSC evaluation.

Evaluation Products' Influence on the Field

Our final measure of influence on the field considers the extent to which the LSC evaluation products were cited throughout the field. The citation analysis uses bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work.

This section describes the 98 products produced by the LSC evaluation and the 187 unique publications (referred to as "citing works") that cited one or more of the LSC evaluation products. Following the product and citing work descriptions, patterns in the citation data are explored through network maps.

Descriptions of the LSC Evaluation Products. The LSC core evaluation produced a total of 98 evaluation related products. Lists of these products and the citing works to each product are presented in Appendix A and B. The products were coded as to their types, fields, and subfields according to the definitions presented below:

Evaluation products were coded by type according to the following:

- Instrument/tools: evaluation instruments (observation protocols, surveys, etc.), instrument manuals, online evaluation tools (logic model tool, resource database);
- Presentations: conference presentations, presentations to NSF or other audiences;
- Publications: journal articles, books, book chapters, published monographs, dissertations, newsletters;
- Reports: evaluation reports, briefing papers, fact sheets, brochures.

Product fields were broadly defined according to the following:

- STEM education/research: general topics or research related to STEM education;
- STEM evaluation: topics or findings connected to a specific STEM evaluation;
- Education—general: education areas/topics outside of STEM education;
- Evaluation—general: evaluation areas/issues outside of STEM evaluation;
- Other: items not falling into any of the above categories.

Products were coded as to the specific sub-field within education or evaluation according to the following:

- Professional development: teacher training during service, including the effects of professional development on student achievement, etc.;
- Workforce development: high tech, STEM workforce training/development;
- Partnerships/systemic reform: university, school, community partnerships, or state or district-wide systemic reform initiatives;

- Curriculum/instruction/assessment: related to classroom teaching (i.e., instruction methods, curriculum, pedagogy, student assessment, standards-based instruction, math/science reform);
- Evaluation: evaluation-related topics, such as multi-site evaluation theory or other general evaluation theory or knowledge, including evaluation instruments, protocols, and manuals[11];
- Materials development: development of instructional materials;
- Project capacity building: issues related to program improvement, sustainability, advisory committee development, dissemination, replication;
- Other fields.

LSC Evaluation Product Types. Not quite half of the 98 products the LSC core evaluation produced were evaluation reports (n=42), while publications (n=33) comprised about a third of the products. Additionally, 13 evaluation instruments and 10 presentations were produced by the core evaluation team (see Table 16).

Table16. Types of products produced by the LSC core evaluation

Product Type	N	%
Report	42	42.9
Publication	33	33.7
Instrument/Tool	13	13.3
Presentation	10	10.2
Total	*98*	*100.0*

As expected, the LSC core evaluation's products were primarily focused on evaluation-related issues connected to the core evaluation (coded as STEM evaluation). Almost 85% of the products were STEM evaluation-related, while 14% were related to STEM education issues and one addressed a general topic in evaluation (see Table 17).

Table 17. Fields of evaluation products

Product Fields	N	%
STEM evaluation	83	84.7
STEM education/research	14	14.3
Evaluation - general	1	1.0
Total	*98*	*100.0*

As shown in Table 18, almost three quarters of the products produced by the LSC core evaluation were related to topics in teacher professional development (n=72). There were also a number of evaluation-related works, including evaluation instruments, produced by the core evaluation team (n=17). To a lesser extent, LSC evaluation products addressed issues in curriculum, instruction, assessment (n=5), partnerships or systemic reform (n=3), and teacher training (n=1).

Table 18. Specific fields of evaluation products

Product Subfields	N	%
Professional Development	72	73.5
Evaluation	17	17.3
Curriculum/Instruction/Assessment	5	5.1
Partnerships/systemic reform	3	3.1
Teacher training/Pre-service	1	1.0
Total	*98*	*100.0*

Citations to the LSC Core Evaluation Products. Thirty-eight of the ninety-eight LSC evaluation products, or almost 40% of the products, received at least one citation. As shown in Table 19, among the 38 products that were cited, approximately one third received only one citation, another third received between two and five citations, and the top third received six or more citations.

Table 19. Number of citations per product

Number of citations	Number of Products	%
0	60	61.2
1	12	12.2
2	6	6.1
3	4	4.1
4	2	2.0
5	1	1.0
6	3	3.1
7	2	2.0
8	1	1.0
9	1	1.0
10	1	1.0
11	1	1.0
23	1	1.0
25	1	1.0
33	1	1.0
48	1	1.0
Total	*98*	*100.0*

Four of the ninety-eight LSC evaluation products stood out from the others in terms of citation levels. These "highly cited" products received between 23 and 48 citations each. The four products, presented in Table 20 below, consisted of two LSC evaluation instruments/manuals and two evaluation reports. While three of the products were produced in the late 1990s, one was published as recently as 2006.

Table 20. Highly-cited LSC evaluation products

Product	Citations	Date	Type	Field	Subfield
Horizon Research, Inc. (1999-2005). *Local Systemic Change through Teacher Enhancement Classroom Observation Protocol*	48	1999	Instrument/ Tool	STEM evaluation	Evaluation
Banilower, E. R., Boyd, S. E., Pasley, J. D., & Weiss, I. R. (2006). *Lessons from a decade of mathematics and science reform: A capstone report for the local systemic change through teacher enhancement initiative.* Chapel Hill, NC: Horizon Research, Inc.	33	2006	Report	STEM evaluation	Professional Development
Weiss, I. R., Montgomery, D. L., Ridgway, C. J., & Bond, S. L. (1998). *Local systemic change through teacher enhancement: Year three cross site report.* Chapel Hill, NC: Horizon Research, Inc.	25	1998	Report	STEM evaluation	Professional Development
Horizon Research, Inc. (1997). *Local Systemic Change 1997-98 Core Data Collection Manual.* Chapel Hill, NC: Author.	23	1997	Instrument/ Tool	STEM evaluation	Evaluation

No significant differences were found in the mean number of citations received by products of different fields or different subfields. However, there were significant differences found in the mean number of citations for different types of evaluation products ($F=5.612$, $df=3$, $p=.001$). The differences in product type accounted for 15.2% of the variance in mean citations. As shown in Table 21, LSC evaluation instruments were cited most frequently, averaging over 8 citations per instrument. Post-hoc analyses showed that instruments had a significantly higher mean citation rate than reports, publications, or presentations.

Table 21. Citations for types of products

Evaluation Product Type	Number of products	Number of Citations	Mean Citations/ Product
Instrument	13	111	8.54
Report	42	126	3.00
Publication	33	11	.33
Presentation	10	0	.00
Total	*98*	*248*	*2.53*

Description of the Works Citing the LSC Evaluation Products. A total of 248 references were found to the 98 evaluation products produced by the LSC evaluation. The 248 references came from 187 unique citing works, which are detailed in Appendix B. While the vast majority of the citing works cited only one product (n=155), twenty one of the citing works cited two evaluation products (11%), seven cited three LSC evaluation products (4%), and the remaining four cited more than three products (see Table 22).

The publication that cited 14 LSC evaluation products was an LSC evaluation product itself. This was the LSC capstone report detailing findings and lessons learned from the years the LSC core evaluation was funded (Banilower, Boyd, Pasley, & Weiss, 2006). The publication that cited seven LSC products was an external publication: the Council of Chief State School Officers' "Study of Math and Science Professional Development Programs: Improving Evaluation of Professional Development" (2006). The publication that cited five LSC products was also not affiliated with the evaluation field. This journal article entitled "Influence of a shared leadership model in creating a school culture of inquiry and collegiality" was written by Khourey-Bowers, C., Dinko, R.L., and Hart, R.G., and published in the Journal of Research in Science Teaching in 2005.

Table 22. Number of references to LSC evaluation products by citing works

Number of LSC references per citing work	N	%
1	155	82.9
2	21	11.2
3	7	3.7
4	1	.5
5	1	.5
7	1	.5
14	1	.5
Total	*187*	*100.0*

As shown in Table 23, almost 60% of the citations to LSC core evaluation products (n=108) came from sources not affiliated with the LSC core evaluation, any of the other three evaluations under consideration in the Beyond Evaluation Use study (i.e., ATE, CETP, MSP-RETA), or a source affiliated with NSF or related STEM projects and staff[12].

After non-affiliated sources, works authored by individuals affiliated with the LSC core evaluation (n=21) and works authored by individuals affiliated with one of the Local Systemic Change projects (n=17) were most frequent. These citation counts also show that the LSC core evaluation was influential on Collaboratives for Excellence in Teacher Preparation (CETP) projects around the country (n=9) as well as the CETP core evaluation (n=6). Many of these citations were to the LSC classroom observation protocol which served as a model for several

CETP projects and the CETP core evaluation. Other areas where the LSC core evaluation products had influence was on other NSF projects (n=9), Math-Science Partnership (MSP) projects (n=5), NSF organizational sources (n=5), and PIs and staff members affiliated with either the ATE, CETP, or MSP-RETA evaluations who were writing in contexts other than the four specific multi-site evaluations under consideration in the Beyond Evaluation Use grant project.

The citing works were predominantly US in origin (published or presented.) There were five citations that came from international sources, in terms of either their publication or presentation: two from Canada, and one each from Australia, Europe, and Puerto Rico.

Table 23. Authors of citing works

Citing Work Authors	N	%
Non-Affiliated Source	108	57.8
LSC Core Evaluation	21	11.2
LSC Project	17	9.1
CETP Project	9	4.8
NSF - Other Project	9	4.8
CETP Core Evaluation	6	3.2
MSP Project	5	2.7
NSF Source	5	2.7
Evaluation PI	5	2.7
Evaluation Staff	2	1.1
Total	*187*	*100.0*

Citing works were categorized by type according to the same categorization scheme used with the products, with the addition of codes for "electronic resources" which was used for Web pages, archived listserv messages, or similar online sources, and the category "other." As shown in Table 24, publications were the most common source for citations to LSC products (39%), followed by evaluation reports (27%), and to a lesser extent electronic resources (13%), presentations (12%), and evaluation instruments or tools (3.7%). There were ten other types of citing works, including five NSF announcements (such as RFPs, media advisory, and program announcements), three grant proposals, and two course syllabi.

Table 24. Citing work types

Types	N	%
Publication	72	38.5
Report	50	26.7
Electronic Resource	25	13.4
Presentation	23	12.3
Other	10	5.3
Instrument/Tool	7	3.7
Total	*187*	*100.0*

As shown in Table 25, over 80% of works citing LSC core evaluation products were related to STEM education and research (n=96) or STEM evaluation (n=61). A smaller minority of the citing works were related to educational issues outside the STEM areas (n=25). Only a few works were related to general evaluation issues (n=4), and one work was related generally to the fields of physics and astronomy.

Table 25. Citing work fields

Fields	N	%
STEM education/research	96	51.3
STEM evaluation	61	32.6
Education – general	25	13.4
Evaluation	4	2.1
Other	1	.5
Total	*187*	*100.0*

The citing works represented a variety of specific fields within education and evaluation. Two-thirds of the citing works discussed issues in teacher professional development (n=74) and curriculum, instruction, and assessment (n=52). This makes sense given the nature of the LSC program. Evaluation-related publications were also prominent (n=25), including those publications related to evaluation instruments. A number of citing works were also related to systemic reform initiatives at the state or district level (n=15). As enumerated in Table 26, there were a number of other areas that the LSC evaluation products reached, including teacher training, workforce development, materials development, data-based decision making, and school administration and leadership.

Table 26. Citing work sub-fields

Subfields	N	%
Professional Development	74	39.6
Curriculum/Instruction/Assessment	52	27.8
Evaluation	25	13.4
Partnerships/systemic reform	15	8.0
NSF	6	3.2
Other	5	2.7
Teacher training/Pre-service	4	2.1
Workforce Development	3	1.6
Materials Development	2	1.1
Project Capacity Building	1	.5
Total	*187*	*100.0*

Citation Patterns of LSC Core Evaluation Products. While the data presented above convey one picture of the citations to LSC core evaluation products, this section presents a graphical picture of the interconnected web of LSC-related citations. For the following images, the program NetDraw was used to create maps of the networks of LSC evaluation products and citations. In each of the maps, arrows point from the citing work to the evaluation product it is citing. The products and citing works are in the same position in each of the maps; only the colors change to highlight different characteristics of interest to illustrate citation patterns.

Figure 1 presents the overall citation structure of LSC evaluation products (blue squares) and works authored by individuals other than LSC core evaluation staff or consultants (red circles).

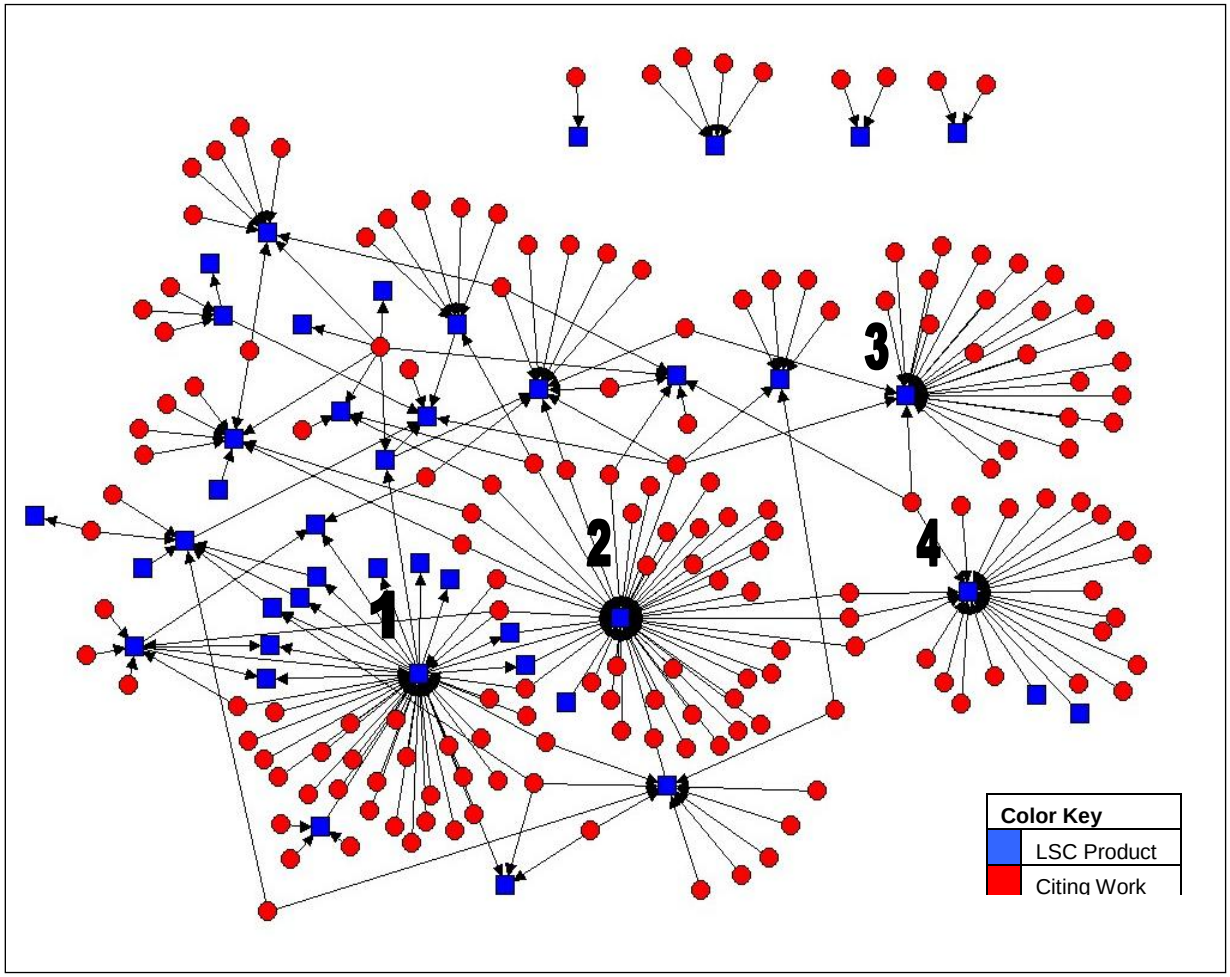


Figure 1. LSC evaluation products and citations

As shown in Figure 2, there is widespread citing by non-LSC core authors, as well as evidence of internal citing between LSC evaluation products. Although there are a few LSC evaluation products that are cited by one or more citing works in isolation from the rest of the products, the majority of the products and citations are connected together through networks of co-citation. There are relatively few instances of products that cite more than one or two LSC products, however. The majority of works cite one product only.

There are two central hubs in the network that show high levels of interconnectivity. The first, toward the left side of the figure and indicated by the number "1," is Banilower, et al. (2006) "Lessons from a decade of mathematics and science reform: A capstone report for the local systemic change through teacher enhancement initiative." In addition to being cited 33 times, this product itself cited 14 other LSC evaluation products. The second hub, in the center of the figure and indicated by the number "2," is the "Local Systemic Change through Teacher Enhancement Classroom Observation Protocol" which is cited a total of 48 times by both

internal and external sources. A close inspection of the citations to and from these two hub products shows a high degree of interconnectedness between these two products and other LSC products and citations. There are two other highly cited LSC evaluation products marked in the figure. The "Year-Three Cross Site Report" is marked with the number "3." The "Core Data Collection Manual" is marked with the number "4." While both of these products are highly cited, they are less highly interconnected with other citations and products than the other two highly cited works.

In Figure 2, the colors on the image have changed, although the LSC evaluation products remain blue squares and non-affiliated citing works are red. The remainder of the citing works has been colored to illustrate the affiliation of the authors to any of programs under consideration in the Beyond Evaluation Use project, to another NSF funded project, or to the NSF organization.

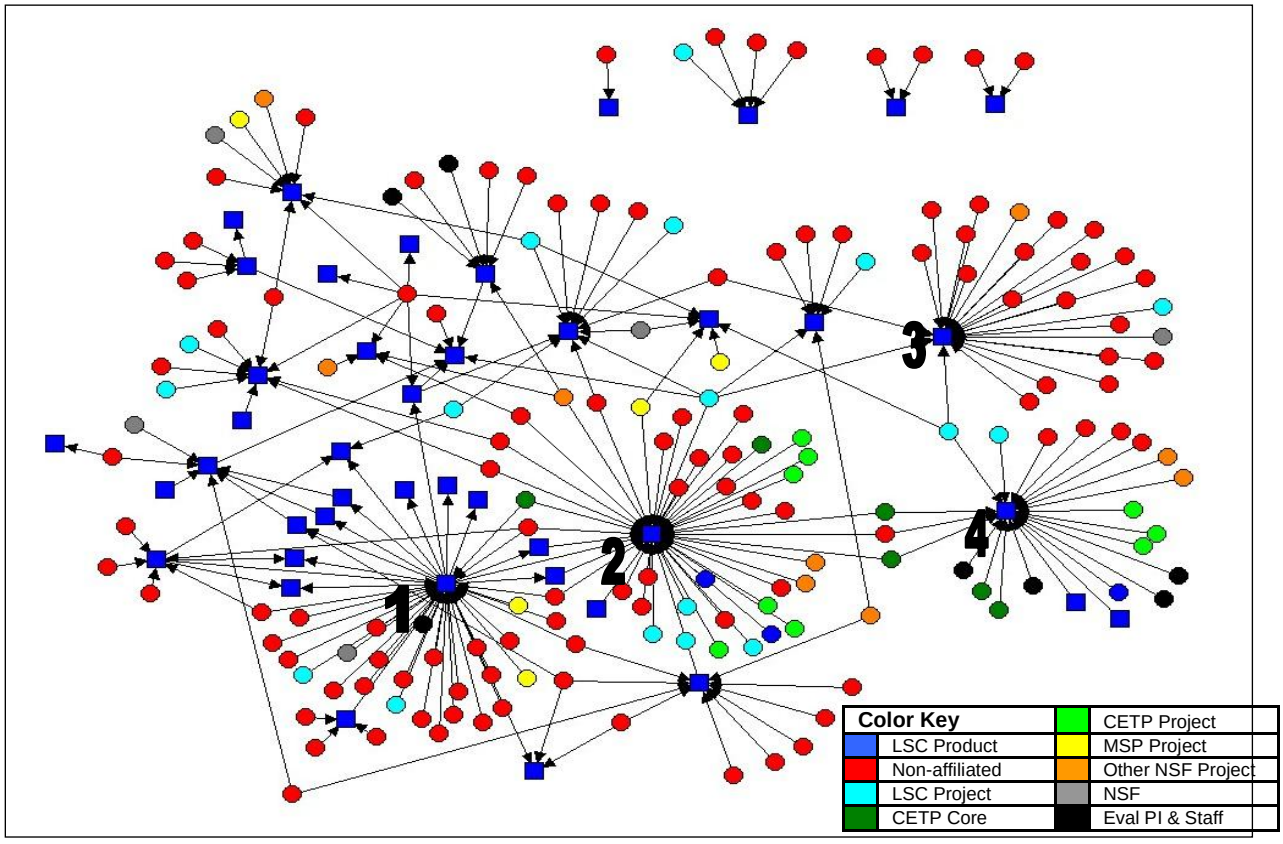


Figure 2. Authors of products and citing works

As shown in Figure 3, most of the evaluation products were cited by a variety of different fields within STEM education. This is evident with three of the four highly cited products. The Classroom Observation Protocol (#2), Year-Three Cross Site Report (#3), and Core Data Collection Manual (#4) are all cited by a number of professional development-related works, but also reached a variety of other fields. The "Lessons Learned" article (#1), on the other hand, appears to be grounded in the professional development and curriculum, instruction, and assessment literature. Not only are the vast majority of the LSC products that it cited related to

professional development, the majority of its citing works are from the professional development and curriculum fields, with a handful of citations from works addressing systemic reform.

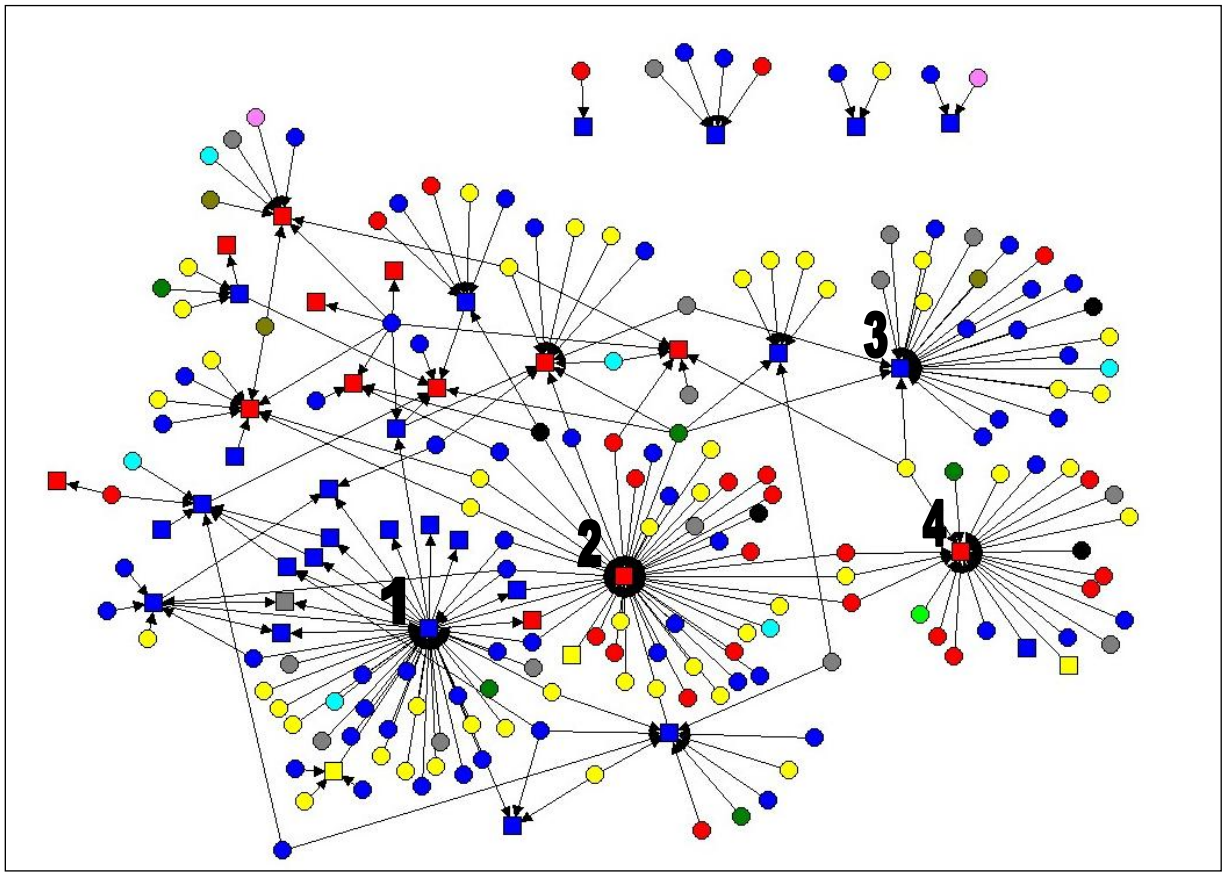


Figure 3. Fields of products and citing works

Color Key			
	Professional Development		Teacher training
	Curriculum/Instruct/Assess.		Materials Development
	Evaluation		Project Capacity Building
	Partnerships/Systemic Reform		Workforce Development
	NSF		Other

Citation Analysis Discussion and Conclusions. The LSC core evaluation was conducted by a private research company rather than coming out of the "publish or perish" culture of academia. NSF, however, was interested in maximizing the impact of the evaluation's findings and stressed dissemination during the no-cost extension period of the ten-year grant. Today, almost thirteen years since the evaluation was first funded, about 40% of the core evaluation's products have been cited at least once and a number of products have been cited widely within the fields of professional development, curriculum, instruction, and assessment, and evaluation.

The analyses presented above suggest that two types of products were most highly influential on the STEM fields. The first type of highly cited product is evaluation instruments such as the Classroom Observation Protocol, Core Data Collection Manual, and the surveys of teachers and administrators. The interest in these instruments can likely be attributed to two factors. First, the instruments filled a need within the field; there was a lack of high-quality instruments for measuring outcomes of interest in mathematics and science teacher professional development programs. Second, the LSC program evaluation involved data collection from 88 LSC projects in 467 districts across the United States. Consequently, a high number of teachers, administrators, and evaluators were involved with assessing or being assessed by these instruments.

The second type of product that was highly cited is summary oriented reports that present the most important findings and lessons learned from the evaluation. These included such publications as the projects recently published capstone report "Lessons from a decade of mathematics and science reform" and other key evaluation reports that presented summative findings.

While a number of evaluation instruments, reports, and publications were found to have an impact on the STEM fields, it is important to note that none of the numerous LSC Core Evaluation newsletters or presentations made by HRI staff was found to be cited in this analysis. While these dissemination efforts may have contributed indirectly to a greater knowledge of the core evaluation and its findings, these types of products appear to be relatively ineffective in contributing to a more formalized spread of knowledge as captured by citations.

Influence on the Field Summary

As supported by the data presented in the preceding sections, the LSC core evaluation appears to have had limited influence on the fields of STEM education and evaluation. While over 90% of the interviewed project personnel reported that the LSC core evaluation had some level of impact on both the STEM education and the evaluation communities, the remainder of the data collected as part of this study support a less enthusiastic characterization of the LSC core evaluation's influence on the fields of STEM education and evaluation.

The field study and the survey of journal editors revealed that evaluations in general were not viewed as influential. In keeping with that overall finding, few respondents to the NSF PI survey were able to name an evaluator or publication related to the LSC core evaluation. Two of the survey respondents reported having been involved in LSC; and, when asked to identify an influential evaluator, one individual identified an evaluator affiliated with the LSC core evaluation. Because of the small number of responding PIs that reported having been involved in the LSC core evaluation, it was not possible to determine if there was any relationship between influence and involvement.

To gather more information about influence on the field, the research team conducted a three-question survey of STEM education and evaluation journal editors. One of the three editors that responded to the journal editor survey named one STEM education evaluator who was the PI of the LSC evaluation.

The citation analysis showed that 24 LSC core evaluation products had been referenced, with the instruments noted as the most influential. Approximately 40% of the core evaluation's products were cited at least once in fields such as professional development, curriculum, instruction, and assessment, and evaluation.

Overall, while the LSC project personnel surveyed perceived relatively high levels of influence, the NSF PI survey and the citation analysis, while showing evidence of some influence, did not detect a high level of influence of the LSC core evaluation on the fields of stem education and evaluation.

Overall Summary: Relationship Between Involvement and Use

The LSC core evaluation was intended to assess the status of individual LSC projects, as well as to collect information from as many diverse projects that could be aggregated to inform broader messages to NSF, Congress and the public. NSF was the primary 'client' for the evaluation and therefore the projects and the fields of STEM education and evaluation were secondary audiences, not directly the intended users.

Each LSC project was required to have a local evaluator who collected data using standard questionnaires and protocols designed by the core evaluation team. Early on, a limited number of project personnel were involved in providing input as the core team designed the instruments and protocols. Throughout the evaluation, the core team involved all project PIs and evaluators in required meetings that provided orientation and training on the use of the standardized data collection instruments.

This case study showed a modest relationship between project involvement in the LSC core evaluation and the influence of the evaluation's processes and products on the projects. It should be noted that project-level involvement was not the only variable affecting influence, and the effect of involvement was confounded by a wide range of perceptions among survey respondents and interviewees about what constitutes involvement. For some, in depth involvement in one activity was seen as constituting substantial involvement, while for others participation in several different activities was necessary to feel the substantial involvement. It appeared from the surveys and interviews with project personnel, however, that perceptions of involvement were positively linked to perceptions of use. Other variables affecting use included where the project was in its life cycle, the perceived relevance of the core evaluation to the local project, and interactions with NSF.

The LSC core evaluation had only a small discernable effect on the fields of STEM education and evaluation. This effect was probably diffused and not directly attributable to the LSC evaluation. The analysis of the instances where LSC core evaluation products were cited in scholarly research showed that there was some use, particularly of the instruments, by people in LSC projects and elsewhere. In addition, the survey of NSF PIs revealed that the respondents had little knowledge of evaluations in general and practically no knowledge of the LSC evaluation.

In summary, the LSC core evaluation had a moderate level of influence on the LSC projects, which was enhanced by the involvement of the projects in the evaluation. The effect on the fields of STEM education and evaluation was much less apparent in the data collection conducted for this case study.

Appendix A: LSC Core Evaluation Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC001	Banilower, E. R. & Germuth A. A. (2004). Structural equation models for LSC teacher questionnaire and classroom observation data. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Evaluation	1	LSC003
LSC002	Banilower, E. R. & Shimkus, E. S. (2004). Professional development observation study. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC003	Banilower, E. R., Boyd, S. E., Pasley, J. D., & Weiss, I. R. (2006). Lessons from a decade of mathematics and science reform: A capstone report for the local systemic change through teacher enhancement initiative. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	33	1026, 1030, 1042, 1042, 1044, 1045, 1054, 1071, 1077, 1094, 1106, 1115, 1125, 1145, 1148, 1155, 1157, 1160, 1163, 1168, 1170, 1183, 1186, 1215, 1223, 1226, 1228, LSC045, 1238, LSC046, 1246, 1261, 1273
LSC004	Banilower, E. R., Rosenberg, S. L., & Weiss, I. W. (2006). Local systemic change through teacher enhancement: 2003–06 cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC005	Banilower, E.R. (2000). Local systemic change through teacher enhancement: A summary of project efforts to examine the impact of the LSC on student achievement. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	3	LSC003, 1180, LSC045
LSC006	Banilower, E.R. (2002). Results of the 2001–2002 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	2	LSC003, 1209
LSC007	Bowes, A. S. & Banilower, E. R. (2004). LSC classroom observation study: An analysis of data collected between 1998 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC008	Boyd, S. E., Banilower, E. R., Pasley, J. D., & Weiss, I.R. (2004). Local systemic change initiative capstone report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC009	Boyd, S. E., Banilower, E. R., Pasley, J. D., & Weiss, I. R. (2003). Progress and pitfalls: A cross-site look at local systemic change through teacher enhancement. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	10	1002, 1003, 1022, 1042, 1048, 1095, 1209, 1240, 1245, 1253
LSC010	Crawford, R. A. & Banilower, E. R. (2004). LSC Principal Questionnaire Study: A cross-sectional analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC011	Flora, D. B. & Panter, A. T. (1999). Analysis of the psychometric structure of the LSC surveys (Technical Report). Chapel Hill, NC: University of North Carolina at Chapel Hill.	Report	STEM evaluation	Evaluation	7	LSC043, LSC043, LSC014, 1110, 1217, LSC031, LSC033
LSC012	Germuth, A. A. & Banilower, E. R. (2004). Results of the 2002–03 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC013	Germuth, A. A., Banilower, E. B., & Shimkus, E. S. (2003). Test-retest reliability of the local systemic change teacher questionnaire. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Evaluation	1	LSC043
LSC014	Heck, D. J. & Crawford, R. A. (2004). LSC teacher questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	2	LSC003, 1029
LSC015	Heck, D. J. & Crawford, R. A. (2004). LSC teacher questionnaire study: Indicators of systemic change, a longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Partnerships/ systemic reform	2	LSC003, LSC045
LSC016	Heck, D. J., Rosenberg, S. L., & Crawford, R. A. (2006). LSC teacher questionnaire study: A longitudinal analysis of data collected between 1997 and 2006. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC017	Horizon Research, Inc. & Westat, Inc. (2001). Revised handbook for studying the effects of the LSC on students. Rockville, MD: Westat, Inc.	Report	STEM evaluation	Evaluation	1	1152
LSC018	Horizon Research, Inc. (1996). Local systemic change through teacher enhancement: 1995 cross-site technical report. Chapel Hill, NC: Author.	Report	STEM evaluation	Professional Development	0	
LSC019	Horizon Research, Inc. (2001). What have we learned? Local systemic change initiatives share lessons from the field. Chapel Hill, NC: Author.	Report	STEM evaluation	Professional Development	2	1116, 1165
LSC020	Horizon Research, Inc. (2005). Designing large scale education initiatives: Lessons from the local systemic change through teacher enhancement program. Chapel Hill, NC: Author.	Report	STEM education/ research	Professional Development	0	
LSC021	Horizon Research, Inc. (2006) Local systemic change through teacher enhancement: Foundations monograph. Chapel Hill, NC: Author.	Report	STEM evaluation	Professional Development	0	
LSC022	Pasley, J. (2002). The role of instructional materials in professional development: Lessons learned from the LSC community. From http://www.horizon-research.com/LSC/news/ pasley2002.php.	Report	STEM education/ research	Professional Development	3	LSC003, 1209, 1253
LSC023	Rosenberg, S. L., Heck, D. J., & Banilower, E. R. (2005). Does content preparation moderate the impacts of professional development? A longitudinal analysis of LSC teacher questionnaire data. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM education/ research	Professional Development	0	
LSC024	Shimkus, E. S. & Banilower, E. R. (2004). LSC project ratings study: An analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC025	Shimkus, E. S. & Banilower, E. R. (2004). LSC teacher interview study: An analysis of data collected between 1999 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC026	Shimkus, E.S., & Banilower, E.R. (2004, June). Local Systemic Change: Comparison of Teachers' and Principals' Perceptions of Factors Influencing Instruction. Chapel Hill, NC: Horizon Research Inc.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment	0	

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC027	Simpson, M. A. & Banilower, E. R. (2004). LSC principal questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC028	Simpson, M. A. & Banilower, E. R. (2004). Results of the 2003–04 Study of the Impact of the Local Systemic Change Initiative on Student Achievement in Science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC029	Weiss, I. R. (2006). Designing large scale education initiatives: Lessons from the local systemic change through teacher enhancement program. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM education/ research	Professional Development	0	
LSC030	Weiss, I. R., Arnold, E. E., Banilower, E. R., & Soar, E. H. (2000). Local systemic change through teacher enhancement: Year five cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC031	Weiss, I. R., Arnold, E. E., Banilower, E. R., & Soar, E. H. (2001). Local systemic change through teacher enhancement: Year six cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	7	LSC006, LSC012, 1152, 1158, LSC027, LSC028, 1245
LSC032	Weiss, I. R., Banilower, E. R., & Shimkus, E. S. (2004). Local systemic change through teacher enhancement: Year nine cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	1222
LSC033	Weiss, I. R., Banilower, E. R., Crawford, R. A., & Overstreet, C.M. (2003). Local systemic change through teacher enhancement: Year eight cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	6	1050, 1107, 1111, 1120, 1131, 1251
LSC034	Weiss, I. R., Banilower, E. R., Overstreet, C. M., & Soar, E. H. (2002). Local systemic change through teacher enhancement: Year seven cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	6	1040, 1110, 1178, 1191, 1240, 1275
LSC035	Weiss, I. R., Gellatly, G. B., Montgomery, D. L., Ridgway, C. J., Templeton, C. D., & Whittington, D. (1999). Executive summary of the local systemic change through teacher enhancement: Year four cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	4	1080, 1081, 1134, 1156

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC036	Weiss, I. R., Gellatly, G. B., Montgomery, D. L., Ridgway, C. J., Templeton, C. D., & Whittington, D. (1999). Local systemic change through teacher enhancement: Year four cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	2	1076, 1164
LSC037	Weiss, I. R., Montgomery, D. L., Ridgway, C. J., & Bond, S. L. (1998). Local systemic change through teacher enhancement: Highlights of the year three cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC038	Weiss, I. R., Montgomery, D. L., Ridgway, C. J., & Bond, S. L. (1998). Local systemic change through teacher enhancement: Year three cross site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	25	1006, 1019, 1020, 1024, 1037, 1038, 1056, 1066, 1072, 1073, 1105, 1110, 1135, 1138, 1162, 1173, 1177, 1190, 1198, 1202, 1207, 1234, 1254, 1264, 1276
LSC039	Weiss, I. R., Rapp, K. A., & Montgomery, D. L. (1997). Local systemic change through teacher enhancement: Highlights of the year two cross site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC040	Weiss, I. R., Rapp, K. A., & Montgomery, D. L. (1997). Local systemic change through teacher enhancement: Executive summary of the year two cross site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC041	Zhang, X. & Frechtling, J. (2005) Preliminary meta-analysis for local systemic change student outcome studies. Rockville, MD: Westat	Report	STEM evaluation	Professional Development	0	
LSC042	Zhang, X. & Wang, L. (2002). National science foundation local systemic change: Evaluation of data collection and analysis for cohort 1–4 projects on student achievement. Rockville, MD: Westat.	Report	STEM evaluation	Professional Development	2	LSC003, LSC045

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC043	Banilower, E. R., Heck, D. J., & Weiss, I. R. (2007). Can professional development make the vision of the standards a reality? The impact of NSF's local systemic change through teacher enhancement initiative. Journal of Research in Science Teaching, 44(3), 375–395.	Publication	STEM education/ research	Professional Development	3	1011, 1208, 1221
LSC044	Heck, D. J., Banilower, E. B., & Weiss, I. R. (Accepted for publication). Studying the effects of effective professional development: The case of NSF's local systemic change through teacher enhancement program. Journal for Research in Mathematics Education.	Publication	STEM education/ research	Professional Development	0	
LSC045	Weiss, I. R. & Pasley, J. D. (2006, March). Scaling up instructional improvement through teacher professional development: Insights from the local systemic change initiative. CPRE Policy Briefs, RB-44.	Publication	STEM education/ research	Professional Development	5	1030, 1043, 1106, 1166, 1267
LSC046	Weiss, I. R. & Pasley, J. D. (2007). Teaching math and science: Improving instruction through local systemic change initiatives. Phi Delta Kappan, 88(9), 669–675.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment	3	1004, 1060, 1085
LSC047	Weiss, I.R. (2005). Evaluation of the nsf local systemic change program: Implications for preparing new teachers of mathematics. Association of Mathematics Teacher Educators (AMTE) Connections Vol. 15, Issue 1.	Publication	STEM evaluation	Teacher training/Pre-service	0	
LSC048	Horizon Research, Inc. (1999-2005). Local Systemic Change through Teacher Enhancement Classroom Observation Protocol	Instrument/ Tool	STEM evaluation	Evaluation	48	1010, LSC001, LSC003, 1018, 1031, 1035, 1046, 1055, 1061, 1065, 1074, 1078, 1096, 1097, 1098, 1098, 1099, 1100, 1101, 1103, 1106, 1110, 1125, CETP013, CETP012, CETP007, 1131, 1136, 1140, 1141, 1175, 1187, 1188, 1197, 1199, 1200, 1214, 1216, 1218, 1219, 1225, 1227, 1241,

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
						LSC066, 1261, LSC007, LSC009, 1268
LSC049	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Mathematics 6-12 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	4	1018, 1029, 1131, 1196
LSC050	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement "Teacher Questionnaire" = not specific as to which teacher questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	11	1032, 1047, 1049, 1073, LSC014, 1110, 1136, 1146, 1159, 1180, 1255 ,
LSC051	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Mathematics K-8 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	1	1029
LSC052	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Principal Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	6	1029, 1032, 1138, 1159, 1200, 1248
LSC053	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Professional Development Observation Protocol	Instrument/ Tool	STEM evaluation	Evaluation	8	1029, 1032, 1041, 1161, 1172, 1176, 1220, 1278
LSC054	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Science 6-12 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	1	1029
LSC055	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Science K-8 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	9	1029, 1036, 1065, 1119, 1176, 1195, 1199, LSC065, 1258
LSC056	Horizon Research, Inc. (2000-2005). Local Systemic Change through Teacher Enhancement Teacher Interview Protocol	Instrument/ Tool	STEM evaluation	Evaluation	0	
LSC057	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Teacher Interview Protocol: Final Year	Instrument/ Tool	STEM evaluation	Evaluation	0	
LSC058	Horizon Research, Inc. (2000-2005). Local Systemic Change through Teacher Enhancement Teacher Interview Summary	Instrument/ Tool	STEM evaluation	Evaluation	0	
LSC059	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Teacher Interview Summary:	Instrument/ Tool	STEM evaluation	Evaluation	0	

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
	Final Year					
LSC060	Horizon Research, Inc. (1997). Local Systemic Change 1997-98 Core Data Collection Manual. Chapel Hill, NC: Author.	Instrument/ Tool	STEM evaluation	Evaluation	23	1033, LSC096, 1088, 1089, 1090, 1091, 1092, 1103, 1104, 1112, 1118, CETP013, CETP012, CETP011, 1138, 1150, 1154, 1192, LSC026, 1230, 1231, 1274, CETP010
LSC061	Briars, Diane & Kaufman, Mark. (2005). Professional Development Strategies for Mathematics Improvement. Paper presented at the NSDC Annual Conference.	Presentation	STEM education/ research	Professional Development	0	
LSC062	Weiss, Iris R. (2007, Feb.) How Do You Know that You are Making Progress? Evaluating Your Professional Development Program. Paper presented at the 2007 Leadership Seminar on Mathematics Professional Development, Portland OR, February14-17, 2007.	Presentation	Evaluation - general	Professional Development	0	
LSC063	Weiss, Iris R. (2006, April). Issues in Designing Professional Development for Teachers of Mathematics. Paper presented at the 2006 NCSM Meeting.	Presentation	STEM education/ research	Professional Development	0	
LSC064	Smith, P. Sean. (2005, June). Effective Professional Development in Science: Why Do We Need it? What Does It Look Like? Presentation at the National LASER K-8 Science Education Strategic Planning Institute	Presentation	STEM education/ research	Professional Development	0	
LSC065	Weiss, Iris R. (2005, May). Findings from the LSC. Presentation at the CCSSO Conference.	Presentation	STEM evaluation	Professional Development	0	
LSC066	Weiss, Iris R. (2005, November). High Quality Mathematics and Science Instruction: Where do we stand? How do we move forward? Presentation to Pittsburgh Public Schools.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
LSC067	Weiss, Iris R. (2005, April). LSCs, MSPs, SSIs, etc.: Implications of Research on Federally-Funded Projects for Improvement of Mathematics Education District-Wide. Presentation at the	Presentation	STEM evaluation	Partnerships/ systemic reform	0	

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
	2005 NCSM Meeting.					
LSC068	Weiss, Iris R. (2005, October). LSCs, MSPs, SSIs, etc.: Implications of Research on Federally-Funded Projects for Improvement of Mathematics Education District-Wide. Presentation to the North Carolina Supervisors.	Presentation	STEM evaluation	Partnerships/ systemic reform	0	
LSC069	Weiss, Iris R. (2004, November). Linking Science and Literacy in the Classroom: The National Context. Presentation at the NSTA Science and Literacy Conference.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
LSC070	Weiss, Iris R. (2005, November). Research to Practice: Applying What We Know To Improve Mathematics and Science Education. Presentation to the Network Connections Conference.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
LSC071	LSC Core Update Newsletter, September 2004	Publication	STEM evaluation	Professional Development	0	
LSC072	LSC Core Update Newsletter, June 2004	Publication	STEM evaluation	Professional Development	0	
LSC073	LSC Core Update Newsletter, March 2004	Publication	STEM evaluation	Professional Development	0	
LSC074	LSC Core Update Newsletter, December 2003	Publication	STEM evaluation	Professional Development	0	
LSC075	LSC Core Update Newsletter, September 2003	Publication	STEM evaluation	Professional Development	0	
LSC076	LSC Core Update Newsletter, June 2003	Publication	STEM evaluation	Professional Development	0	
LSC077	LSC Core Update Newsletter, March 2003	Publication	STEM evaluation	Professional Development	0	
LSC078	LSC Core Update Newsletter, December 2002	Publication	STEM evaluation	Professional Development	0	
LSC079	LSC Core Update Newsletter, September 2002	Publication	STEM evaluation	Professional Development	0	
LSC080	LSC Core Update Newsletter, June 2002	Publication	STEM	Professional	0	

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
			evaluation	Development		
LSC081	LSC Core Update Newsletter, March 2002	Publication	STEM evaluation	Professional Development	0	
LSC082	LSC Core Update Newsletter, December 2001	Publication	STEM evaluation	Professional Development	0	
LSC083	LSC Core Update Newsletter, September 2001	Publication	STEM evaluation	Professional Development	0	
LSC084	LSC Core Update Newsletter, July 2001	Publication	STEM evaluation	Professional Development	0	
LSC085	LSC Core Update Newsletter, March 2001	Publication	STEM evaluation	Professional Development	0	
LSC086	LSC Core Update Newsletter, September 2000	Publication	STEM evaluation	Professional Development	0	
LSC087	LSC Core Update Newsletter, July 2000	Publication	STEM evaluation	Professional Development	0	
LSC088	LSC Core Update Newsletter, March 2000	Publication	STEM evaluation	Professional Development	0	
LSC089	LSC Core Update Newsletter, April 1999	Publication	STEM evaluation	Professional Development	0	
LSC090	LSC Core Update Newsletter, September 1999	Publication	STEM evaluation	Professional Development	0	
LSC091	LSC Core Update Newsletter, November 1999	Publication	STEM evaluation	Professional Development	0	
LSC092	LSC Core Update Newsletter, May 1998	Publication	STEM evaluation	Professional Development	0	
LSC093	LSC Core Update Newsletter, August 1998	Publication	STEM evaluation	Professional Development	0	
LSC094	LSC Core Update Newsletter, November 1998	Publication	STEM evaluation	Professional Development	0	
LSC095	LSC Core Update Newsletter, January 1997	Publication	STEM evaluation	Professional Development	0	
LSC096	LSC Core Update Newsletter, March 1997	Publication	STEM evaluation	Professional Development	0	

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC097	LSC Core Update Newsletter, May 1997	Publication	STEM evaluation	Professional Development	0	
LSC098	LSC Core Update Newsletter, November 1996	Publication	STEM evaluation	Professional Development	0	

Appendix B: Citing Works to LSC Core Evaluation Products

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1002	Abell, S., Ehlert, M., Lannin, J., Marra, R., Cole, J., Park Rogers, M., Lee, M., Wang, C. & Musikul, K. (2005, April). Profiles of Mathematics and Science Teacher Professional Development Programs: Theoretical Basis, Activities, and Effectiveness. Paper presented at the annual meeting of the American Educational Research Association, Montreal, Quebec.	Presentation	STEM education/ research	Professional Development
1003	Abell, S.K, Lannin, J.K., Marra, R.M., Ehlert, M.W., & Cole, J.S. (2007). Multi-site evaluation of science and mathematics teacher professional development programs: The project profile approach. Studies in Educational Evaluation, 33(2), 135-158.	Publication	STEM evaluation	Evaluation
1004	Access My Library. (2007). Teaching math and science: Improving instruction through Local Systemic Change initiatives. Available from: http://www.accessmylibrary.com/coms2/summary_0286-31239646_ITM	Electronic Resource	STEM evaluation	Professional Development
1006	Alabama Mathematics, Science, and Technology Initiative Committee. (2000, October). Report on the review of the literature. Presented to the Alabama State Board of Education.	Report	STEM education/ research	Partnerships/ Systemic reform
1010	Arizona Collaborative for the Excellence of Preparation of Teachers. (nd). About RTOP. Available from: http://physicsed.buffalostate.edu/AZTEC/RTOP/RTOP_full/about_RTOP.html	Report	STEM evaluation	Evaluation
1011	Astrophysics Data System. (nd). Digital Library for Physics and Astronomy. Available from: http://www.adsabs.harvard.edu/	Electronic Resource	Other	Other
1018	Basista, B., & Mathews, S. (2002). Integrated Science and Mathematics Professional Development Programs. School Science and Mathematics, 102(7), 359.	Publication	STEM education/ research	Professional Development
1019	Beasley, W. (2000). Systemic Reform through Teacher Change: congruence and contradiction. Teaching Education, 11(1), 39.	Publication	Education - general	Partnerships/ Systemic reform

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1020	Center for Science, Mathematics, and Engineering Education. (1999). Designing Mathematics or Science Curriculum Programs: A Guide for Using Mathematics and Science Education Standards. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1022	Chrispeels, J. (2004). Learning to lead together: The promise and challenge of sharing leadership. Thousand Oaks, CA: Sage Publications Inc	Publication	Education - general	Other
1024	Commission of the European Communities. (2007). Accompanying document to the proposal for a recommendation of the European Parliament and of the Council on the quality of teacher education. Brussels: author.	Report	Education - general	Teacher training/Pre-service
1026	Connecticut Department of Higher Education. (2006, September). 2007 Teacher Quality Partnership Grant Program: Request for Proposals and Application Guidelines. Hartford, CT: author.	Other	Education - general	Partnerships/ Systemic reform
1029	Council of Chief State School Officers. (2006, September 9). Study of Math and Science Professional Development Programs: Improving Evaluation of Professional Development. Washington, DC: author. Available from: www.ccsso.org/content/PDFs/Eval_Tools_Chart-Change_Instruction.xls	Electronic Resource	STEM evaluation	Professional Development
1030	Council of Chief State School Officers. (2007, November). Resources for State Leaders and Evaluators: Improving Evaluation of Professional Development. Washington, DC: CCSSO. Available from http://www.ccsso.org/projects/improving_evaluation_of_professional_development/Resources/	Electronic Resource	STEM evaluation	Professional Development
1031	Czerniak, C.M. & Beltyukova, S. (2003, July). Tapestries: A successful elementary science curriculum in metropolitan Toledo. Urban Affairs Center: The University of Toledo.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1032	Czerniak, C.M. (2003). Christa McAuliffe Award: 2003 Final Proposal Submission for the Toledo Area Partnership in Education: Support Teachers as Resources to Improve Elementary Science (TAPESTRIES). Available from: http://www.aascu.org/programs/mcauliffe/pdf/toledo.pdf	Other	STEM education/ research	Curriculum/ Instruction/ Assessment
1033	D'Ambrosio, B. S., Boone, W.J., & Harkness, S.S. (2004, January). Planning District-Wide Professional Development: Insights Gained From Teachers and Students Regarding Mathematics Teaching in a Large Urban District. School Science and Mathematics, 104(1), 5-15.	Publication	STEM education/ research	Professional Development
1035	Danin, S. (2000, August 11). Re: Classroom Observation Protocol. Message posted on the Discussion: Intensive Sites Task Force for E=MSC2 listserv. Available from: http://www.edgateway.net/cs/ege/forum/cs_disc/573?x-showcontent=message_text.	Electronic Resource	STEM evaluation	Professional Development
1036	Davidson, A.B. (2000, November). Eighth Grade Science Teachers Use of Instructional Time: Comparing Questions from the Third International Mathematics and Science Study (TIMSS) and National Science Foundation Questionnaires. Paper presented at the Annual Meeting of the Mid-South Educational Research Association (Bowling Green, KY, November 15, 2000). Eric document# ED448056.	Presentation	STEM evaluation	Curriculum/ Instruction/ Assessment
1037	Desimone, L.A., Smith, T.M., & Ueno, K. (2006). Are teachers who need sustained, content-focused professional development getting it? An administrator's dilemma. Educational Administration Quarterly, 42(2), 179-215.	Publication	Education - general	Professional Development
1038	Desimone, L.M., Porter, A.C., Garet, M.S., et al. (2002). Effects of professional development on teachers' instruction: Results from a three-year longitudinal study. Educational Evaluation and Policy Analysis, 24(2), 81-122.	Publication	Education - general	Professional Development
1040	Donovan, M.S., & Pellegrino, J.W. (2004). Learning and Instruction: A SERP Research Agenda. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1041	Duggan-Haas, D. Using The Secondary Teaching Analysis Matrix in Professional Development. Presentation available from: people.colgate.edu/ddugganhaas/STAM/STAM%20NSTA%20Workshop.ppt	Presentation	Education - general	Professional Development
1042	Committee on Science Learning, Kindergarten through Eighth Grade, Richard A. Duschl, Heidi A. Schweingruber, and Andrew W. Shouse, Editors. (2007). Taking Science to School: Learning and Teaching Science in Grades K-8. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1043	Education Commission of the States Education Policy Issue Site: Mathematics. (2007). Selected Research and Readings. Available from: http://www.ecs.org/html/IssueSection.asp?issueid=81&s=Selected+Research+%26+Readings	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1044	Education Commission of the States. (nd). Document Number: 6937. Available from http://www.ecs.org/html/Document.asp?chouseid=6937.	Report	Education - general	Professional Development
1045	Educational Research Associates. (2006, April). Successes, Challenges Cited in Math, Science Reform. E-bulletin, 33(15).	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1046	Ellis, M.W., Malloy, C.E., Meece, J.L., & Sylvester, P.R. (2007). Convergence of observer ratings and student perceptions of reform practices in sixth-grade mathematics classrooms. Learning Environments Research, 10(1), 1-15.	Publication	STEM evaluation	Curriculum/ Instruction/ Assessment
1047	Eun, B. & Heining-Boynton, A.L. (2007, Sept.). Impact of an English-as-a-Second-Language Professional Development Program. The Journal of Educational Research, 101(1), 36-49.	Publication	Education - general	Professional Development
1048	Evaluation of Professional Development Projects. (nd). Resources. Missouri: author. Available from: http://www.pdeval.missouri.edu/resources_2.html	Electronic Resource	Evaluation	Professional Development

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1049	Ferrini-Mundy, J. (2004). SIRG Brief 3: What Does It Mean to Be Standards-based? Issues in Conceptualizing, Measuring, and Studying Alignment with Standards. In Frank K. Lester, Jr., and Joan Ferrini-Mundy (Eds.) Proceedings of the National Council of Teachers of Mathematics Research Catalyst Conference.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1050	Fletcher, C., Meyer, J.D., Barufaldi, J.P., Lee, E., Tinoca, L., & Bohman, T.M. (2004, March). The Science Classroom Profile: Design, Development, and Use. Paper presented at the National Association of Research in Science Teaching conference, Vancouver, Canada.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1054	Fraser, B. J. & Kahle, J. B. (2007). Classroom, Home and Peer Environment Influences on Student Outcomes in Science and Mathematics: An analysis of systemic reform data. International Journal of Science Education, 1-19.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1055	Frechtling, J. (2001, February 9). Responses to Questions from LSC Conference Participants. Paper presented at the Studying the Effects of the LSC on Students Conference.	Presentation	STEM evaluation	Professional Development
1056	Garet, M.S., Porter, A.C., Desimone, L., et al. (2001). What makes professional development effective? Results from a national sample of teachers. American Educational Research Journal, 38(4), 915-945.	Publication	Education - general	Professional Development
1060	Goliath Business Knowledge on Demand. (2007). Teaching math and science: Improving instruction through Local Systemic Change initiatives. Available from: http://goliath.ecnext.com/coms2/gi_0199-6619640/Teaching-math-and-science-improving.html	Electronic Resource	STEM evaluation	Professional Development
1061	Good, K. (2000, September 18). Re: Classroom Observation Protocol. Message posted on the Discussion: Intensive Sites Task Force for E=MSC2 listserv. Available from: http://www.edgateway.net/cs/ege/forum/cs_disc/573?x-showcontent=message_text.	Electronic Resource	STEM evaluation	Professional Development

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1065	Grable, L.L. & Park, J.C. (2002, February). Anchoring Inquiry And Content For Teachers Through Technology: Middle School Mathematics And Science Integration. In B.S. Wojnowski & D.G. Haase, (Eds). Proceedings of the Conference on K-12 Outreach from University Science Departments. Raleigh, NC: The Science House, North Carolina State University.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1066	Grant, M.L. (2005). Impact of the Kindergarten Teacher Reading Academy on the Instructional Practices of Kindergarten Teachers. Doctoral Dissertation: Mississippi State University.	Publication	Education - general	Curriculum/ Instruction/ Assessment
1071	Hake, R. (2006, March). [Phys-l] Lessons from a Decade of Mathematics and Science Reform. Electronic posting to the Phys-L: Forum for Physics Educators Listserv. Archived at: https://carnot.physics.buffalo.edu/archives/2006/03_2006/msg00005.html	Electronic Resource	STEM education/ research	Professional Development
1072	Hamilton, L.S. & Guarino, C.M. (2004, November). Working Paper: Measuring the Practices, Philosophies, and Characteristics of Kindergarten Teachers. RAND Education Working Paper Series, WR-199-EDU. Prepared for the National Center for Education Statistics by RAND Education, Santa Monica, CA.	Report	Education - general	Curriculum/ Instruction/ Assessment
1073	Hamilton, L.S., McCaffrey, D.F., Stecher, B.M., et al. (2003). Studying large-scale reforms of instructional practice: An example from mathematics and science. Educational Evaluation and Policy Analysis, 25(1), 1-29.	Publication	STEM evaluation	Partnerships/ Systemic reform
1074	Haney, J.J., Lumpe, A.T., Czerniak, C.M. & Egan, V. (2002, August). From Beliefs to Actions: The Beliefs and Actions of Teachers Implementing Change. Journal of Science Teacher Education, 13(3), 171-187.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1076	Hartshorne, R. (2005). Effects of Integrating Hypermedia into Elementary Science Professional Development on Science Content Knowledge. Journal of Science Education and Technology, 14(4), 415.	Publication	STEM education/ research	Professional Development

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1077	Harvard Graduate School of Education. (2006, March/April). WIDE World Email Newsletter. Available from http://learnweb.harvard.edu/wide/news/newsletter/2006_04.cfm	Publication	Education - general	Professional Development
1078	Heck, D. (2001, February). Proposal for Measuring the LSC Program's Effects on Student Achievement in Science. Paper presented at the Studying the Effects of the LSC on Students Conference.	Presentation	STEM evaluation	Professional Development
1080	Heller, J.I., Daehler, K.R., & Shinohara, M. (2003, Fall). Connecting all the Pieces: Mosaic approach to evaluation makes a complete picture. Journal of Staff Development, 24(4).	Publication	Evaluation	Evaluation
1081	Heller, J.I., Daehler, K.R., Shinohara, M., & Kaskowitz, S.R. (2004, April). Fostering Pedagogical Content Knowledge about Electric Circuits Through Case-Based Professional Development. Paper presented at the annual meeting of the National Association for Research on Science Teaching, Vancouver, Canada.	Presentation	STEM education/ research	Professional Development
1085	HighBeam Research. (2007). Teaching math and science: Improving instruction through Local Systemic Change initiatives. Available from: http://www.highbeam.com/doc/1G1-163471125.html	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1088	Huffman, D, & Kalnin, J. (2003, August). Collaborative inquiry to make data-based decisions in schools. Teaching and Teacher Education. 19(6):569-580	Publication	Education - general	Project Capacity Building
1089	Huffman, D. & Lawrenz, F. (2004). The Impact of a State Systemic Initiative on U.S. Science Teachers and Students. Journal International Journal of Science and Mathematics Education, 1(3), 357-377.	Publication	STEM education/ research	Partnerships/ Systemic reform
1090	Huffman, D. (2006, June). Reforming Pedagogy: Inservice Teacher Education and Instructional Reform. Journal of Science Teacher Education, 17(2), 121-136.	Publication	STEM education/ research	Professional Development

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1091	Huffman, D., Goldberg, F., & Michlin, M. (2003). Using computers to create constructivist learning environments: Impact on pedagogy and achievement. Chesapeake, VA: Association for the Advancement of Computing in Education. Available from: http://www.thefreelibrary.com/Using+computers+to+create+constructivist+learning+environments:...-a0111266274	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1092	Huffman, D., Thomas, K. & Lawrenz, F. (2003, December). Relationship Between Professional Development, Teachers' Instructional Practices, and the Achievement of Students in Science and Mathematics. School Science and Mathematics, 103(8), 78-87.	Publication	STEM education/ research	Professional Development
1094	Inverness Research Associates. (2006, November). The Gilbert Elementary Science Program: The Infrastructure Needed to Sustain High Quality Instruction. Inverness, CA: author.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1095	Inverness Research Associates. (2006, June). The Biological Sciences Curriculum Study, National Academy for Curriculum Leadership: Contributions and Lessons Learned: An Evaluation Brief. Inverness, CA: author.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1096	Iona, S. (nd). Summary of Classroom Observation Protocols. Available from: http://www.colorado.edu/physics/phys4810/phys4810_fa06/protocol.html	Electronic Resource	STEM evaluation	Evaluation
1097	Jeanpierre, B.J. (2004). Two Urban Elementary Science Classrooms: The Interplay between Student Interactions and Classroom Management Practices. Education, 124(4), 664.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1098	Johnson, C.C. (2007, August). Whole-School Collaborative Sustained Professional Development and Science Teacher Change: Signs of Progress. Journal of Science Teacher Education, 18(4), 629-661.	Publication	STEM education/ research	Professional Development
1099	Johnson, C.C. (2007, April). Technical, political and cultural barriers to science education reform. International Journal of Leadership in Education, 10(2), 171-190.	Publication	STEM education/ research	Partnerships/ Systemic reform

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1100	Johnson, C.C., Kahle, J.B., & Fargo, J.D. (2006). A study of the effect of sustained, whole-school professional development on student achievement in science. Journal of Research in Science Teaching, 44(6), 775-786.	Publication	STEM education/ research	Professional Development
1101	Johnson, C.C., Kahle, J.B., & Fargo, J.D. (2006). Effective teaching results in increased science achievement for all students. Science Education, 91(3), 371-383. ED# EJ60015.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1103	Jones, M.T. & Eick, C.J. (2007). Providing Bottom-Up Support to Middle School Science Teachers' Reform Efforts in Using Inquiry-Based Kits. Journal of Science Teacher Education.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1104	Jones, M.T., & Eick, C.J. Implementing inquiry kit curriculum: Obstacles, adaptations, and practical knowledge development in two middle school science teachers. Science Education, 91(3), 492-513.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1105	Kanaya, T., Light, D., & Culp, K.M. (2005). Factors Influencing Outcomes from a Technology-Focused Professional Development Program. Journal of Research on Technology in Education, 37(3), 313.	Publication	STEM education/ research	Professional Development
1106	Kansky, B. (nd). Report Summary: Professional Development – Insights from the Local Systemic Change Initiatives. CPRE Policy Report. Philadelphia, PA: Consortium for Policy Research in Education. Available from: http://mathforum.org/kb/servlet/JiveServlet/download/204-1529691-5505263-383119/RSS47.DOC	Report	STEM education/ research	Professional Development
1107	Katzenmeyer, C. & Lawrenz, F. (2006). National Science Foundation Perspectives on the Nature of STEM Program Evaluation. In Huffman, D. & Lawrenz, F. (Eds.), Critical Issues in STEM Evaluation: New Directions for Evaluation, 109. (pp. 7-18).	Publication	STEM evaluation	Evaluation
1110	Khourey-Bowers, C., Dinko, R.L., & Hart, R.G. (2005). Influence of a shared leadership model in creating a school culture of inquiry and collegiality. Journal of Research in Science Teaching, 42(1), 3-24.	Publication	STEM education/ research	Other

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1111	Kimble, L.L., Yager, R.E. & Yager, S.O. (2006). Success of a Professional-Development Model in Assisting Teachers to Change Their Teaching to Match the More Emphasis Conditions Urged in the National Science Education Standards. Journal of Science Teacher Education, 17(3), 309-322.	Publication	STEM education/ research	Professional Development
1112	Klentschy, M.P., & Maruca, P.M. (2006). The administrator's role in supporting science and literacy in classrooms. In R. Douglas, M.P. Klentschy, & K. Worth. (Ed.), Linking Science and Literacy in the K-8 Classroom (pp. 407-427). Arlington, VA: National Science Teachers Association Press.	Publication	STEM education/ research	Other
1115	Laursen, S.L. (2007). Getting Unstuck: Strategies for Escaping the Science Standards Straitjacket. The Astronomy Education Review, 5(1), 162-177.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1116	Lawrence Hall of Science. (2007). Database of FOSS Research on FOSS and Ongoing Projects. University of California: author. Available from: http://www.lawrencehallofscience.org/foss/scope/research/search.php	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1118	Lawrenz, F. & Post, T. (1999, Spring). Local Systemic Change Initiatives in Science and Mathematics. Research/Practice, 7(1). Minneapolis, MN: Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM education/ research	Professional Development
1119	Lawrenz, F. (2002). Minneapolis Local Systemic Change Initiative Core Evaluation Report: 2001-2002 Final Year. Available from: http://tis.mpls.k12.mn.us/sites/5df1b159-7ce3-4aa3-8e71-8e60a7b98e6c/uploads/LSCeval2002_2.pdf	Report	STEM evaluation	Professional Development
1120	Lawrenz, F. (2007). Review of Science Education Program Evaluation. In S.K. Abell & N.G. Lederman (Eds.), The Handbook of Research on Science Education (pp. 943-963). London: Routledge.	Publication	STEM evaluation	Evaluation
1125	Lawrenz, F., Huffman, D. & Gravely, A. (2007). Impact of the collaboratives for excellence in teacher preparation program. Journal of Research in Science Teaching, 44(9), 1348 - 1369.	Publication	STEM education/ research	Professional Development
1131	Lee, M.H, Olson, T.A.,& Scribner, J.P. (2006, May). Exploring Alternative Teacher Certification Policy and Practice through an Examination of Novice Science Teachers. Paper presented at the STEM ACT Conference, Arlington, VA.	Report	STEM education/ research	Teacher training/Pre-service

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1134	Lord, B., & Miller, B. (2000, March). Teacher Leadership: An Appealing and Inescapable Force in School Reform? Newton, MA: Education Development Center.	Report	Education - general	Partnerships/ Systemic reform
1135	Loucks-Horsley, S., & Matsumoto, C. (1999). Research on Professional Development for Teachers of Mathematics and Science: The State of the Scene. School Science and Mathematics, 99.	Publication	STEM education/ research	Professional Development
1136	Loucks-Horsley, S., Love, N., Stiles, K.E., Mundry, S., & Hewson, P.W. (2003). Designing Professional Development for Teachers of Science and Mathematics, 2nd Edition. Thousand Oaks, CA: Corwin Press.	Publication	STEM education/ research	Professional Development
1138	Lumpe, A.T., Czerniak, C.M. & Haney, J.J. (1999, June). Supporting the Implementation of Inquiry-based Elementary Science Programs: Setting the Stage for Local Reform. Electronic Journal of Science Education, 3(4).	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1140	MacIsaac, D., Sawada, D., & Falconer, K. (2001, April). Using the Reformed Teaching Observation Protocol (RTOP) as a Catalyst for Self-Reflective Change in Secondary Science Teaching. Paper presented at the American Educational Research Association annual conference, Seattle, Washington.	Presentation	STEM evaluation	Professional Development
1141	Madfes, T.J. & Rosen, R. (2004, November). Follow-up evaluation of MASTEP [Mathematics and Science Teacher Education Program]: Final Report. San Francisco: WestEd.	Report	STEM evaluation	Teacher training/Pre-service
1145	Marshall, J.E. & Bock, S.C. (2006, November). Science teacher retention: What if… CSP Connection, 7(2), 3-4.	Publication	STEM education/ research	Professional Development
1146	McCaffrey, D.F., Hamilton, L.S., Stecher, B.M., Klein, S.P., Bugliari, D., & Robyn, A. (2001, November). Interactions among Instructional Practices, Curriculum, and Student Achievement: The Case of Standards-Based High School Mathematics. Journal for Research in Mathematics Education, 32(5), 493-517.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1148	Michaels, S. Shouse, A.W., & Schweingruber, H.A. (2007). Ready, Set, Science!: Putting Research to Work in K-8 Science Classrooms. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1150	Milanowski, T. & Kimball, S. (2006). Ed Leadership & Policy Analysis 703: Evaluating and Supporting Quality Classroom Teaching Course Syllabus. Madison, WI: University of Wisconsin.	Other	Evaluation	Curriculum/ Instruction/ Assessment
1152	Millsap, M.A., & O'Reilly, F. (nd). Building Evaluation Capacity with Theory-Based Evaluation. Abt Associates, Inc. Used as an example of a successfully funded grant proposal on the NSF REC website. Available from: http://www.ehr.nsf.gov/rec/programs/evaluation/	Other	Evaluation	Evaluation
1154	Morrell, P.D., Flick, L., & Wainwright, C. (2004, May). Reform Teaching Strategies Used by Student Teachers. School Science and Mathematics, 104(5), 199-213.	Publication	STEM education/ research	Teacher training/Pre-service
1155	MSPnet: Math Science Partnerships' Learning Network. (2007). Library: Lessons from a Decade of Mathematics and Science Reform: A Capstone Report for the Local Systemic Change Through Teacher Enhancement Initiative. Available from http://hub.mspnet.org/index.cfm/12760	Electronic Resource	STEM evaluation	Professional Development
1156	Mundry, S. (2005, June). What Experience Has Taught Us About Professional Development. National Network of Eisenhower Regional Consortia and Clearinghouse.	Report	STEM education/ research	Professional Development
1157	National Education Writers Association. (2006, June). Education Reform: No Child Left Behind and Science Education.	Publication	STEM education/ research	Partnerships/ Systemic reform
1158	National Science Foundation, Directorate for Education and Human Resources. (2002, January). Program Solicitation: Instructional Materials Development program. Washington, DC: author. http://nsf.gov/pubs/2002/nsf02067/nsf02067.txt	Other	STEM education/ research	NSF
1159	National Science Foundation. (1998, May 28). Agency information collection activities: Cross Projects Evaluation of the Local Systemic Change Through Teacher Enhancement Program. Federal Register, 63(102). Available from: http://vlex.com/vid/23373036	Other	STEM evaluation	NSF
1160	National Science Foundation. (2006, March). Media Advisory 06-006 Panel Reviews a Decade of Math and Science Education Reform. Available from http://www.nsf.gov/news/news_summ.jsp?cntn_id=106713.	Other	STEM education/ research	NSF

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1161	National Science Foundation: Division of Elementary, Secondary and Informal Education (ESIE). (2003, May 11). Response to the Committee of Visitors (COV) Report Presidential Awards for Excellence in Mathematics and Science Teaching (PAEMST) Program. Available from: http://www.nsf.gov/od/oia/activities/cov/ehr/2003/Response_PAEMSTcov.doc.	Report	STEM evaluation	NSF
1162	National Science Foundation; Directorate for Education and Human Resources; Division of Elementary, Secondary, and Informal Education. (2000). Program Announcement: Centers for Learning and Teaching program. Washington DC: National Science Foundation.	Other	STEM education/ research	NSF
1163	National Science Teachers Association. (2006, May). NSTA Position Statement: Professional Development in Science Education. Downloaded 10/29/07 from http://www.nsta.org/about/positions/profdev.aspx.	Electronic Resource	STEM education/ research	Professional Development
1164	National Science Teachers Association. (nd). TE-MAT: Teacher Education Materials Project: A Database for K-12 Mathematics and Science Professional Development Providers. Available at: http://www.te-mat.org/ConceptualFramework/references.aspx	Electronic Resource	STEM education/ research	Materials Development
1165	National Sciences Resources Center. (2007). Professional Development Research. Washington, DC: The National Academies. Available from: http://www.nsrconline.org/professional_development/research.html	Electronic Resource	Education - general	Professional Development
1166	National Staff Development Council. (2007, November 20). Learning Communities. Available from: http://64.78.6.92/standards/learningcommunities.cfm?printPage=1&	Electronic Resource	Education - general	Professional Development
1168	Needham Public Schools. (2007, June). K-8 Science Program Review Report: Phase II. Needham, MA: author.	Report	STEM evaluation	Partnerships/ Systemic reform
1170	Nelson, G.D., & Landel, C.C. (2006). A collaborative approach for elementary science. Educational Leadership, 64(4), 72-75.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

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1172	North Cascades and Olympic Science Partnership. (2003, June 2). Response to Concerns/Questions. Available from: ncosp.wwu.edu/projectgoals/ReviewResponsesFinal.doc	Report	STEM evaluation	Partnerships/ Systemic reform
1173	Paek, P.L, Ponte, E., Sigel, I., Braun, H., & Powers, H. (2004). A Portrait of Advanced Placement Teachers' Practices. College Board Research Report No. 2005-7; ETS RR-05-09. Princeton, NJ: Educational Testing Service.	Report	Education - general	Curriculum/ Instruction/ Assessment
1175	Piburn, M., & Sawada, D. (2001). Reformed Teaching Observation Protocol (RTOP): Reference Manual. ACEPT Technical Report No. IN00-3. Arizona State University: Arizona Collaborative for Excellence in the Preparation of Teachers.	Instrument/Tool	STEM evaluation	Evaluation
1176	Poole, S.J., deGrazia, J.L., & Sullivan, J.F. (1999, November). Assessing K-12 pre-engineering outreach programs. Paper presented at the 29th Annual Frontiers in Education Conference, San Juan: Puerto Rico.	Presentation	STEM evaluation	Workforce Development
1177	Porter, A.C., Garet, M.S., Desimone, L., Yoon, K.S., & Birman, B.F. (2000). Does Professional Development Change Teaching Practice? Results from a Three-Year Study. Washington DC: US Department of Education.	Report	STEM education/ research	Professional Development
1178	Post, T.R., Davis, J.D., Maeda, Y., Edwin, A.C., Andersen, E., Kahan, J.A., & Harwell, M.R. (2004, April). Standards-based Mathematics Curricula and Middle Grade Students' Performance on Standardized Achievement Tests. Minneapolis, MN: University of Minnesota.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1180	Raghavan, K., Cohen-Regev, S., & Strobel, S.A. (2001). Student Outcomes in a Local Systemic Change Project. School science and mathematics, 101(8), 417.	Publication	STEM education/ research	Professional Development
1183	Rhode Island Board of Governors for Higher Education. (2006). PK-16 Advisory Committee on Science Report. Rhode Island: author.	Report	STEM education/ research	Partnerships/ Systemic reform
1186	Rose, M.A. (2007, January). ITEDU 694: Curriculum Development in Technology Education: RESOURCES. Syllabus for course taught at Ball State University.	Other	STEM education/ research	Curriculum/ Instruction/ Assessment

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1187	Sawada, D., Piburn, M.D., Judson, E., Turley, J., Falconer, K., Benford, R., & Bloom, I. (2002). Measuring Reform Practices in Science and Mathematics Classrooms: The Reformed Teaching Observation Protocol. School Science and Mathematics, 102 (6), 245.	Publication	STEM evaluation	Evaluation
1188	Sawada, D., Piburn, M., Turley, J., Falconer, K., Benford, R., Bloom, I., & Judson, E. (2000, March). Reformed Teaching Observation Protocol (RTOP): Training Guide. ACEPT Technical Report No. IN00-2. Arizona State University: Arizona Collaborative for Excellence in the Preparation of Teachers.	Report	STEM evaluation	Evaluation
1190	Schaetzel, K., Peyton, J.K., & Burt, M. (2007, October). Professional Development for Adult ESL Practitioners: Building Capacity. Washington, DC: Center for Applied Linguistics. Available from: http://www.cal.org/caela/esl_resources/briefs/profdev.html	Electronic Resource	Education - general	Professional Development
1191	Schoen, H.L., Cebulla, K.J,, Finn, K.F., et al. (2003). Teacher variables that relate to student achievement when using a standards-based curriculum. Journal for Research in Mathematics Education, 34(3), 228-259.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1192	Schuster, D., Cobern, W.W., Applegate, B., Schwartz, R., Vellom, P., & Undreiu, A. (2007, October). Assessing pedagogical content knowledge of inquiry science teaching: Developing an assessment instrument to support the undergraduate preparation of elementary teachers to teach science as inquiry. Preprint version. To be published in the Proceedings of the National STEM Assessment Conference on Assessment of Student Achievement, hosted by the National Science Foundation and Drury University.	Presentation	STEM evaluation	Evaluation
1195	Sciulli, J.A. (2004). Teaching Science Through Inquiry in K-5 Classrooms: Analysis of Change in Practice. Doctoral dissertation: Duquesne University.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1196	Scribner, J.P., Akiba, M., Olson, T.A., Lee, M.H., Nivens, R., & Valentine, E. (2007). Mentoring and teacher practice: Understanding organizational support of standards-based mathematics and science instruction. Paper presented at the Annual Convention of the University Council for Educational Administration, Alexandria, VA	Presentation	STEM education/ research	Professional Development
1197	Seeds of Innovation Website. (nd). Pre-Classroom Observation Interview. Available from: www.iaete.org/challenge/ upload/DIAL_Class_Obs_Protocol.pdf	Instrument/Tool	STEM evaluation	Evaluation
1198	Shapley, K., Sheehan, D., Sturges, K., Caranikas-Walker, F., Huntsberger, B., & Maloney, C. (2006, April). Evaluation of the Texas Technology Immersion pilot: First-year results. Austin: TX: Texas Center for Educational Research.	Report	STEM evaluation	Workforce Development
1199	Shattuck, D. (2006). Technology Implementation in the Classroom. Master's Thesis; North Carolina State University.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1200	Shefner, R. (2007, November). Evaluation without Pain: An Improved Matrix. Presented at the Illinois Mathematics and Science Partnerships Collaborative Meeting #2: Core Needs Assessments and Evaluation Design.	Presentation	STEM evaluation	Evaluation
1202	Shymansky, J.A., Yore, L.D., & Anderson, J.O. (2004). Impact of a school district's science reform effort on the achievement and attitudes of third- and fourth-grade students. Journal of Research in Science Teaching, 41(8), 771-790.	Publication	STEM education/ research	Partnerships/ Systemic reform
1207	Smith, T.M., & Desimone, L.M. (2003, Spring). Do Changes in Patterns of Participation in Teachers' Professional Development Reflect the Goals of Standards-based Reform? Educational HORIZONS, 119-129.	Publication	Education - general	Professional Development
1208	Smith, T.M., Desimone, L.M., Zeidner, T.L., et al. (2007). Inquiry-oriented instruction in science: Who teaches that way? Educational Evaluation and Policy Analysis, 29(3), 169-199.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1209	Snow-Renner, R., & Lauer, P.A. (2005). McREL Insights Professional Development Analysis. Denver, CO: Mid-continent Research for Education and Learning.	Report	Education - general	Professional Development

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1214	State Educational Technology Directors Association. (nd). ED Tech Instrument Table. Available from: www.setda.org/c/document_library/ get_file?folderId=4&name=Examples+of+DDDM.xls	Electronic Resource	STEM education/ research	Evaluation
1215	State of Louisiana. (2007-08). Proposals for LaSIP PD for K-12 Teachers. Available from www.lasip.org/nov28docs/CVCfor2007-08PDProjects.2.ppt	Presentation	STEM education/ research	Professional Development
1216	Stazesky, P.B. (2004, April). Measuring the impact of the quality of mathematics instruction on student achievement during the middle school years: Pilot study. Newark, DE: Delaware Education Research & Development Center, University of Delaware.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1217	Supovitz, J.A., & Turner, H.M. (2000). The effects of professional development on science teaching practices and classroom culture. Journal of Research in Science Teaching, 37(9), 963-980.	Publication	STEM education/ research	Professional Development
1218	Taber, M.R., & Quadracci, K. (2006). Building geoscience vocabularies using a data visualization tool (WorldWatcher). In C.A. Manduca and D.W. Mogk, Eds. Earth and Mind: How Geologists Think and Learn about Earth. Special Paper 413. Boulder, CO: The Geological Society of America.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1219	Taylor, J. & Gess-Newsome, J. (2007, January). Exploring Tools and Methods for Measuring Pedagogical Content Knowledge. Presentation at the Annual Meeting of the Association for Science Teacher Education, Clearwater, FL, January 2007.	Presentation	STEM evaluation	Evaluation
1220	The American Association for the Advancement of Science, Northwestern University, The University of Michigan, & Michigan State University. (2006). Preparing the Next Generation of Curriculum Materials Leaders: A Center for the Design, Analysis, and Implementation of Innovative Science Materials. A Proposal to the National Science Foundation for a Center for Learning and Teaching with a Focus on Instructional Materials in Science. Available from: http://ed-web2.educ.msu.edu/CCMS/Webpages/Propo	Other	STEM education/ research	Materials Development
1221	The Art of Teaching Science. (2007, March 6). Reform in Science Teaching, What Does Research Tell Us? Look to a New Cadre. Available from: http://artofteachingscience.org/?cat=6	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment

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1222	Council of Chief State School Officers. (2006, March). Selected Research and Resources: Evaluation tools. Available online at: www.ccsso.org/content/PDFs/SelectedResearchandResourcesAnnotatedBib.doc	Electronic Resource	Education - general	Evaluation
1223	The IDEA Partnership: National Association of State Directors of Special Education. (2007, November). Reports, Guides, Instructional Resources, and Opportunities: Math and Science Combined in One Source. Retrieved from http://www.ideapartnership.org/report.cfm?reportid=185.	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1225	Urban Systemic Program. (2004, October 15). Response to the Committee of Visitors. Available from: nsf.gov/od/oia/activities/cov/ehr/2004/Response_USP_COVRpt.doc	Report	STEM evaluation	NSF
1226	Utah State Office of Education. (nd). Science. Available from http://www.schools.utah.gov/curr/sci/SSECC/2006%2009%2021/LSC%20Capstone%20Reportpdf.pdf	Report	STEM education/ research	Professional Development
1227	Van Zoest, & L.R., Stockero, S.L. (2006). The role of curriculum materials in new teachers' practice. In Alatorre, S., Cortina, J.L., Sáiz, M., and Méndez, A.(Eds) (2006). Proceedings of the 28th annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education. Mérida, México: Universidad Pedagógica Nacional.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1228	Vanosdall, R., Klentschy, M., Hedges, L.V, & Weisbaum, K.S. (2007, April). A Randomized Study of the Effects of Scaffolded Guided-Inquiry Instruction on Student Achievement in Science. Paper presented at the Annual Meeting of the American Educational Research Association, Chicago, Illinois.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1230	Wainwright, C. L., Flick, L. B., Morrell, P. (Fall 2003). Development of instruments for assessment of instructional practices in standards-based teaching. Journal of Mathematics and Science: Collaborative Explorations, 6(1), 21-46.	Publication	STEM evaluation	Evaluation
1231	Wainwright, C., Flick, L., Morrell, P.D., & Schepige, A. (2004). Observation of Reform Teaching in Undergraduate Level Mathematics and Science Courses. School Science and Mathematics, 104(7), 322-336.	Publication	STEM education/ research	Evaluation

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1234	Weingarth, D. (2006). Perceptions of STARR (Select Teachers as Regional Resources) Teachers Concerning their Professional Development Experience. Doctoral dissertation: University of Missouri-Columbia.	Publication	Education - general	Professional Development
1238	Weiss, I.R. (2006, February). Research on Effective Professional Development. Paper presented at the Council of Chief State School Officers conference: Bridging the Gap between Research & Practice in Professional Development.	Presentation	STEM education/ research	Professional Development
1240	Weiss, I.R., Heck, D.J., Miller, B.A., & Cross, K. (2004, June). Handbook for Enhancing Strategic Leadership in the Math and Science Partnerships. Horizon Research, Inc. and Education Development Center.	Report	STEM education/ research	Partnerships/ Systemic reform
1241	Weiss, I.R., Pasley, J.D., Smith, P.S., Banilower, E.R. & Heck, E.R. (2003, May). Looking Inside the Classroom: A Study of K–12 Mathematics and Science Education in the United States. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1245	Westfall, S.A. (2007). The Effect of Professional Development on Physical Education Teachers' Use of Assessment in the Classroom. Doctoral dissertation: Virginia Tech.	Publication	Education - general	Professional Development
1246	National Education Association (2002-2007). What the Research Says. Available from http://www.nea.org/math/research-math.html	Electronic Resource	STEM education/ research	Other
1248	Williams, V.L., Pane, J.F., Tananis, C.A., & Olmsted, S.S. (2005). Math Science Partnership of Southwest Pennsylvania Year Two Evaluation Report. Rand Education Working Paper Series: WR-270-EDU. Santa Monica, CA: Rand Education.	Report	STEM evaluation	Partnerships/ Systemic reform
1251	Yager, R.E. (2005). Implementing the Changes in Professional Development Envisioned by the National Science Education Standards: Where Are We Eight Years Later? In Exemplary Science: Best Practices in Professional Development. RE Yager, Editor. Arlington, VA: National Science Teachers Association Press.	Publication	STEM education/ research	Professional Development

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1253	Yates, S.M. (2006). Primary Teachers' Mathematics Beliefs, Teaching Practices and Curriculum Reform Experiences. Paper presented at the Australian Association for Research in Education conference, Adelaide, Australia.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1254	Yore, L.D., & Treagust, D.F. (2006). Current realities and future possibilities: Language and science literacy-empowering research and informing instruction. International Journal of Science Education, 28(2-3), 291-314.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1255	Young, B.J. & Lee, S.K. (2005). The Effects of a Kit-Based Science Curriculum and Intensive Science Professional Development on Elementary Student Science Achievement. Journal of Science Education and Technology, 14(5/6), 471-481.	Publication	STEM education/ research	Professional Development
1258	Berkeley/Charleston County School District. (2003, January). Report of Fifth Grade Outcome Study, Science for all Students, 2001-2002. Report No. 03-226. Charleston, SC: author.	Report	STEM evaluation	Professional Development
1261	Blank, R.K., de las Alas, N., & Smith, C. (2007, February). Analysis of the Quality of Professional Development Programs for Mathematics and Science Teachers: Findings from a Cross-State Study. Washington DC: Council of Chief State School Officers.	Report	STEM education/ research	Professional Development
1264	Boyle, B., Lamprianou, I, & Boyle, T. (2005). A Longitudinal Study of Teacher Change: What makes professional development effective? Report of the second year of the study. School Effectiveness and School Improvement, 16(1), 1.	Publication	Education - general	Professional Development
1267	Business-Higher Education Forum. (2007). An American Imperative: Transforming the Recruitment, Retention, and Renewal of our Nation's Mathematics and Science Teaching Workforce. Available from: http://www.bhef.com/solutions/quickfacts_061107.pdf	Report	STEM education/ research	Professional Development
1268	Bybee, R.W., Taylor, J.A., Gardner, A., Van Scotter, P., Powell, J.C., Westbrook, A., & Landes, N. The BSCS 5E Instructional Model: Origins, Effectiveness, and Applications. Colorado Springs, CO: Biological Sciences Curriculum Study.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1273	Cavanagh, S. (2006, March 7). NSF Educator-Training Effort Seen as Helpful. Education Week (web only). Available from: http://realizethedream.org/education_news/remote-frame.jsp?base=%5C/education_news/&id=28492117&hostID=28265214 and http://www.edweek.org/ew/articles/2006/03/08/26nsf.h25.html	Publication	STEM education/ research	Professional Development
1274	Center for Applied Research and Educational Improvement. (2006). SSI Impact Study: Project Description. Minneapolis, MN: University of Minnesota. Available from: http://cehd.umn.edu/CAREI/SI/Description.html	Report	STEM evaluation	Partnerships/ Systemic reform
1275	Center for Inquiry in Science Teaching and Learning. (nd). Bibliography. Available from: http://cistl.wustl.edu/Bibliography/index.php?ProgramID=0&showall=1	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1276	Desimone, L.M., Le Floch, K.C., Ansell, S., & Taylor, J. (2005, April). Are We Asking the Right Questions on National Surveys about Teacher's Professional Development? Insights from Cognitive Interviews with Teachers (Working Draft). Presented at the Annual Meeting of the American Education Research Association.	Presentation	STEM evaluation	Evaluation
1278	Poole, S.J., deGrazia, J.L., & Sullivan, J.F. (2001, December). Assessing K-12 pre-engineering outreach programs. Journal of Engineering Education.	Publication	STEM evaluation	Workforce Development
CETP007	Lawrenz, F., Huffman, D., Appledoorn, K., & Michlin, M. (2002, February). CETP Core Evaluation: Field Test Results. Minneapolis, MN: Center for Applied Research and Educational Improvement, University of Minnesota.	Report	STEM evaluation	Evaluation
CETP010	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. K-12 Surveys Handbook, (Jan, 2001) revised September, 2002	Instrument/Tool	STEM evaluation	Evaluation
CETP011	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. Higher Education Survey Handbook, January, 2001. Revised September 2002	Instrument/Tool	STEM evaluation	Evaluation
CETP012	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. Classroom Observation Handbook, January, 2001 Revised September 2002	Instrument/Tool	STEM evaluation	Evaluation

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
CETP013	Lawrenz, F., Huffman, D., & Appeldoorn, K. (2002, September). Handbook for Evaluation of Activities and Assessments. Minneapolis, MN: Collaboratives for Excellence in Teacher Preparation, University of Minnesota.	Instrument/Tool	STEM evaluation	Evaluation
LSC001	Banilower, E. R. & Germuth A. A. (2004). Structural equation models for LSC teacher questionnaire and classroom observation data. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Evaluation
LSC003	Banilower, E. R., Boyd, S. E., Pasley, J. D., & Weiss, I. R. (2006). Lessons from a decade of mathematics and science reform: A capstone report for the local systemic change through teacher enhancement initiative. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC006	Banilower, E.R. (2002). Results of the 2001–2002 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC007	Bowes, A. S. & Banilower, E. R. (2004). LSC classroom observation study: An analysis of data collected between 1998 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC009	Boyd, S. E., Banilower, E. R., Pasley, J. D., & Weiss, I. R. (2003). Progress and pitfalls: A cross-site look at local systemic change through teacher enhancement. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC012	Germuth, A. A. & Banilower, E. R. (2004). Results of the 2002–03 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC014	Heck, D. J. & Crawford, R. A. (2004). LSC teacher questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC026	Shimkus, E.S., & Banilower, E.R. (2004, June). Local Systemic Change: Comparison of Teachers' and Principals' Perceptions of Factors Influencing Instruction. Chapel Hill, NC: Horizon Research Inc.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
LSC027	Simpson, M. A. & Banilower, E. R. (2004). LSC principal questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
LSC028	Simpson, M. A. & Banilower, E. R. (2004). Results of the 2003–04 Study of the Impact of the Local Systemic Change Initiative on Student Achievement in Science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC031	Weiss, I. R., Arnold, E. E., Banilower, E. R., & Soar, E. H. (2001). Local systemic change through teacher enhancement: Year six cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC033	Weiss, I. R., Banilower, E. R., Crawford, R. A., & Overstreet, C.M. (2003). Local systemic change through teacher enhancement: Year eight cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC043	Banilower, E. R., Heck, D. J., & Weiss, I. R. (2007). Can professional development make the vision of the standards a reality? The impact of NSF's local systemic change through teacher enhancement initiative. Journal of Research in Science Teaching, 44(3), 375–395.	Publication	STEM education/ research	Professional Development
LSC045	Weiss, I. R. & Pasley, J. D. (2006, March). Scaling up instructional improvement through teacher professional development: Insights from the local systemic change initiative. CPRE Policy Briefs, RB-44.	Publication	STEM education/ research	Professional Development
LSC046	Weiss, I.R., & Pasley, J.D. (2007). Teaching math and science - Improving instruction through local systemic change initiatives. Phi Delta Kappan, 88(9), 669-675.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
LSC065	Weiss, Iris R. (2005, May). Findings from the LSC. Presentation at the CCSSO Conference.	Presentation	STEM evaluation	Professional Development
LSC066	Weiss, Iris R. (2005, November). High Quality Mathematics and Science Instruction: Where do we stand? How do we move forward? Presentation to Pittsburgh Public Schools.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
LSC096	Horizon Research. (1997, March). LSC Core Update. Chapel Hill, NC: author.	Report	STEM evaluation	Professional Development

Endnotes

[1] Project "leaders" refers to project primary investigators, co-primary investigators, coordinators, directors, administrators, or project consultants.

[2] Project "leaders" refers to project primary investigators, co-primary investigators, coordinators, directors, administrators, or project consultants.

[3] Number of respondents who provided a response directly related to the question or program evaluation. Some responses referred to the survey itself or indicated that they did not have a comment by writing "none," or "no comment;" these types of responses were not considered direct and are not reported in this table

[4] Banilower, ER, Boyd SE, Pasley JD, and Weiss IR. *Lessons from a Decade of Mathematics and Science Reform: A Capstone Report from the Local Systemic Change through Teacher Enhancement Project.* Accessed at http://www.horizon-research.com/reports/2006/capstone.php

[5] Horizon Research Inc. *Findings from a Decade of Local Systemic Change (LSC) Projects Web site.* Retrieved September 28, 2007 from http://www.pdmathsci.net/about/on 11-29-06.

[6] NSF contracted with Horizon Research, Inc. (HRI) to conduct the national evaluation. Under this contract arrangement, NSF was the primary stakeholder and retained the authority to direct the work of the contractor.

[7] Weiss, Iris. PI Reflections, personal communication (paraphrased).

[8] Banilower, ER, Boyd SE, Pasley JD, and Weiss IR. *Lessons from a Decade of Mathematics and Science Reform: A Capstone Report from the Local Systemic Change through Teacher Enhancement Project.* Accessed at http://www.horizon-research.com/reports/2006/capstone.php

[9] A public release data file containing data from 1996-97, the year the teacher questionnaire was stabilized for longitudinal purposes, to 2005-06, the final year of the LSC core evaluation is available at http://www.horizon-research.com/lscprd/

[10] Weiss, Iris. PI Reflections, personal communication (paraphrased).

[11] Note that evaluation instruments, manuals, and protocols were coded as "evaluation" rather than by the topic the instrument collected data about.

[12] Affiliation with the LSC core evaluation, or other NSF evaluations, was made by compare the citing work author names with the lists of names of evaluation project's PIs and evaluators compiled in the Beyond Evaluation Use survey. Additionally, the content of the citing works was scanned for information that would connect the authors to any of the evaluations or the National Science Foundation.